Avoid-or-Compress Words for Reachable Images Missing Four or Five States

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Abstract

Given a set of states S and a nonempty proper subset $A \subset S$, an avoid-or-compress word either omits a state of A from the image of S or reduces the size of that image. We determine the largest possible length of a shortest such word when S is a reachable image missing four or five states. For strongly connected synchronizing automata, the answers are $\min(n-|A|, 11)$ and $\min(n-|A|, 14)$, respectively, and both are attained over a binary alphabet. The lower bounds follow from uniform constructions. For the upper bounds, short-pair graphs and the linear span of short images reduce the problem to finitely many configurations. A computer-assisted classification eliminates these configurations except for one complete bipartite graph, which is excluded by a combinatorial argument.

1 Introduction

A synchronizing automaton admits a word that sends all its states to one state. Bounds on the length of such a reset word often proceed by compressing an image a little at a time. Avoid-or-compress arguments offer a second way to make progress: instead of immediately reducing an image, one may first move it away from a prescribed state or set. The length needed for this step depends on the sizes of both sets. Determining this dependence is the subject of Open Problem 4 in [7].

Let $\mathcal{A} = (Q, \Sigma, \delta)$ be a complete deterministic automaton, with $n = |Q|$. For $\varnothing \neq A \subsetneq S \subseteq Q$, put

$$\mathrm{ac}(S, A) = \min\{|w| : A \not\subseteq Sw \text{ or } |Sw| < |S|\}, \qquad c(S) = \min\{|w| : |Sw| < |S|\}.$$

We write $\mathcal{F}(n, s, r)$ for the maximum of $\mathrm{ac}(S, A)$ over strongly connected synchronizing automata, reachable images $S = Qv$ of size s, and subsets $A \subset S$ of size r. The corresponding maximum over binary automata is denoted by $\mathcal{F}_{\mathrm{bin}}(n, s, r)$. Throughout, $1 \le r < s \le n$, and we set $m = n - s$ and $q = n - r$.

The avoid-or-compress lemma of Szykuła [6, 7] and the Frankl–Pin bound [2, 5] imply

$$\mathcal{F}_{\mathrm{bin}}(n, s, r) \le \mathcal{F}(n, s, r) \le \min\{q, \tbinom{m+2}{2}\}. \tag{1}$$

Related sharpness questions for avoiding a fixed state were studied in [1]. Here we determine the extrema for all dense images with $m = 4$ or $m = 5$.

Theorem 1. *For every admissible n, r,*

$$\mathcal{F}_{\mathrm{bin}}(n, n-4, r) = \mathcal{F}(n, n-4, r) = \min(n-r, 11)$$

and

$$\mathcal{F}_{\mathrm{bin}}(n, n-5, r) = \mathcal{F}(n, n-5, r) = \min(n-r, 14).$$

Our proof has three components. First, a two-block binary construction attains a linear bound in the number of missing states; smaller parameter ranges are covered by related constructions. Second, we prove the compression bounds $c(S) \le 11$ for $(n-4)$-sets when $n \ge 11$ and $c(S) \le 14$ for $(n-5)$-sets when $n \ge 16$. These bounds hold for arbitrary subsets, so the theorem also remains true if reachability of S is dropped from the definition of the extrema. Third, the upper bounds combine structural lemmas with a finite computation: short-pair graphs locate the omitted states, rank growth bounds the relevant independent-set families, and compatible letter actions are tested jointly. The one exceptional configuration has core $K_{4,4}$ and is ruled out by an overlap argument.

Section 2 records the basic tools and notation. Section 3 constructs the binary examples. Sections 4 and 5 give the finite reduction and its computer-assisted verification, and Section 6 completes the proof of Theorem 1. The technical graph lemmas and the codimension-three compression argument are proved in the appendices.

2 Preliminaries

Words act on the right, and $|w|$ denotes length over Σ. The alphabet is finite but otherwise unrestricted. Subsets are not assumed reachable unless this is stated. An automaton is strongly connected if every state reaches every other state, and synchronizing if $|Qw| = 1$ for some word w. For a subset R, let χ_R be its characteristic row vector. The transition matrix M_w is defined by $M_w(x, y) = 1$ when $xw = y$ and 0 otherwise; thus $M_{uv} = M_u M_v$. The row $\chi_R M_w$ counts trajectories with multiplicity, and has support Rw. A word is *safe for* (S, A) if it neither compresses S nor omits a state of A.

Notation	Meaning
n, s, r	$\lvert Q\rvert, \lvert S\rvert, \lvert A\rvert$
m, q	$n-s$, $n-r$
$c(S), \mathrm{ac}(S, A)$	shortest compression and avoid-or-compress lengths
P_h	graph of pairs merged within h letters
K_h, J_h	nonisolated and isolated vertices of P_h
$\alpha(G), \tau(G)$	independence and vertex-cover numbers
$\mathcal{D}$	a family of independent sets, represented by rows
W, J	finite support and common isolated set
$M_w, d(w)$	transition matrix and rank defect $n - \lvert Qw\rvert$
C_j	maximum $c(R)$ over all $(n-j)$-sets in the given automaton

Lemma 2 (Szykuła [6, 7])**.** *If $A \not\subseteq Sz$ for some word z, then* $\mathrm{ac}(S, A) \le n - |A|$.

Proof. Write $r = |A|$, $s = |S|$, and let V_i be the span over $\mathbb{R}$ of $\chi_S M_w$ for $|w| \le i$. Suppose all words of length at most $n-r$ are safe. Safety gives $|Sw| = |S|$, so $S \to Sw$ is injective: no trajectories collide, and every image state has multiplicity 1. Their rows therefore have total s and value 1 at every coordinate of A. Hence V_{n-r} lies in

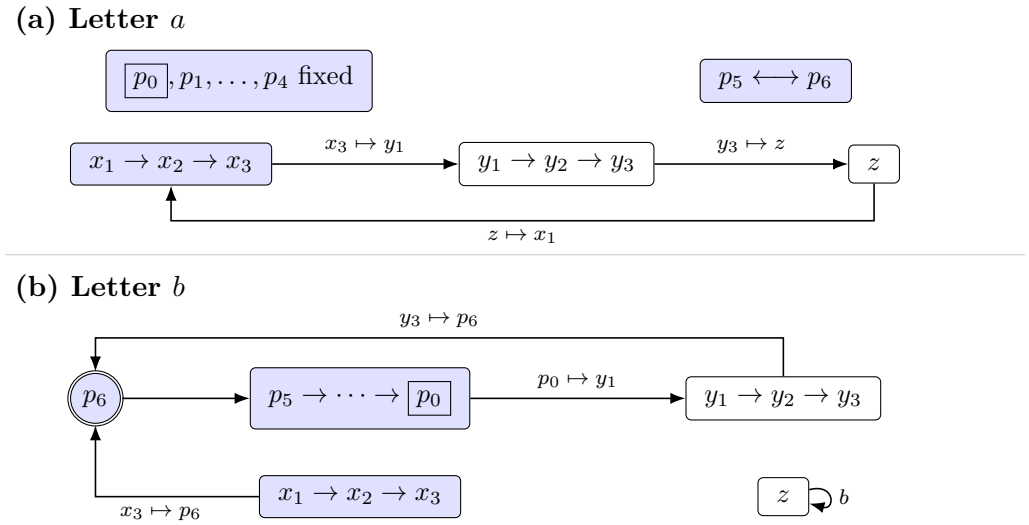
$$H = \{v \in \mathbb{R}^Q : v_a = s^{-1} \textstyle\sum_{x \in Q} v_x \text{ for every } a \in A\}.$$

The r equations are independent since $r < s$, so $\dim H = n - r$. Now $\dim V_0 = 1$, and $V_i = V_{i+1}$ implies invariance under every letter and stabilization forever. Therefore some equality must occur before index $n - r$, or $\dim V_{n-r} \geq n - r + 1$. Stabilization puts $\chi_S M_z$ in H, contradicting its zero coordinate in A and total s. □

In a strongly connected synchronizing automaton with $n \geq 2$, a reset word followed by a path to a state outside A gives the required z. This gives the first bound in (1). The second follows from the Frankl–Pin compression theorem.

3 Binary lower bounds

The construction below separates two obstructions: a pair potential delays compression, while a coloring delays avoidance. Figure 1 shows a representative example.



Blue: initially occupied $S = P \cup X$. Boxed p_0: protected. Double circle: merger target.

Figure 1: The two letter actions for $t = 3$, $p = 7$, $r = 1$. Blue marks the initial image, boxes mark protected states, and the double circle marks the merging target.

3.1 Uniform construction and proof

Theorem 3. *Put $m = n - s \geq 3$ and $q = n - r \geq 2m + 2$. There is a binary strongly connected synchronizing automaton with reachable S attaining $\mathrm{ac}(S, A) = \min\{q, 3m - 1\}$. Consequently $\mathcal{F}_{\mathrm{bin}} = \mathcal{F} = q$ whenever $2m + 2 \leq q \leq 3m - 1$.*

We construct the automaton as follows. Write $t = m - 1 \geq 2$ and $p = n - 2t - 1 \geq r + 3$. Take $P = \{p_0, \ldots, p_{p-1}\}$, $X = \{x_1, \ldots, x_t\}$, $Y = \{y_1, \ldots, y_t\}$ and one further state z. Letter a interchanges the last two P-states, fixes the others, and cycles $x_1, \ldots, x_t, y_1, \ldots, y_t, z$. Letter b sends p_0 to y_1, p_i to p_{i-1} for $i \geq 1$, increments indices in X, Y below t, sends both x_t, y_t to p_{p-1}, and fixes z. Put $S = P \cup X$, $A = \{p_0, \ldots, p_{r-1}\}$ and $L = 3t + 2$.

Lemma 4 (Pair potential)**.** *In the automaton above, $c(S) \geq L$.*

Proof. For $\rho_i = \min\{i, p-2\}$ define a symmetric potential Φ on unordered pairs, zero on singleton pairs, by the following table. Indices in X, Y run from 1 to t, and indices in P from 0 to $p-1$.

pair	potential
P–P, P–X, X–X (distinct)	L
$\{x_i, y_j\}$, $j = i$	$t - i + 1$
$\{x_i, y_j\}$, $j > i$	$2t - i + 1$
$\{x_i, y_j\}$, $j < i$	$L - j$
$\{x_i, z\}$	$2t - i + 1$
$\{y_i, y_j\}$, $i < j$	$L - j$
$\{y_i, z\}$	$2t + 1$
$\{p_i, y_j\}$	$\min\{L, \max\{L - j, 2t + 3 + \rho_i\}\}$
$\{p_i, z\}$	$\min\{L, 2t + 2 + \rho_i\}$

Substitution in the two letter actions gives

$$\Phi(E) \leq 1 + \Phi(Ea), \qquad \Phi(E) \leq 1 + \Phi(Eb).$$

At the boundaries of the index ranges, the inequalities follow as follows. The last-two-P interchange preserves ρ; a downward P-step decreases it by at most one. For P–Y pairs the a-boundary $y_t \to z$ decreases the second term from $2t + 3 + \rho$ to $2t + 2 + \rho$. The b-boundary $p_0 \to y_1$ with $j < t$ has source $L - j$ and target $L - j - 1$, since $j \leq t - 1$; with $j = t$ its target is at least $L - 1$. For X–Y, crossing x_t under a gives Y–Y with potential $L - j - 1$ when $j < i$; crossing under b gives P–Y of at least that potential. The kernel pair $\{x_t, y_t\}$ has potential one. Crossing y_t in a Y–Y pair under a gives Y–z of potential $2t + 1 = L - t - 1$. For X–z, the a-boundary is $\{x_t, z\} \to \{y_1, x_1\}$, with values $t + 1, t$; for Y–z, every a-image has potential $2t$. All remaining cases either retain potential, decrement an index by one, or enter a type whose potential is at least the source minus one. Clipping at L cannot increase a positive difference.

Every distinct pair in S has potential L. Telescoping along a merging word proves its length is at least L; every compression merges an initial pair, giving the claimed lower bound.

□

Lemma 5 (Color cells)**.** *Every word of length less than q satisfies $A \subseteq Sw$.*

Proof. Use singleton P-cells and t color cells, giving x_i, y_i color i and z color 1. Let T_a swap the last two P-cells and cycle the colors, and let T_b cycle $p_0, 1, \ldots, t, p_{p-1}, \ldots, p_1$. Let $C_{x,B} = 1$ if state x belongs to cell B, and 0 otherwise. Write T_a, T_b also for the corresponding permutation matrices. Both letters satisfy

$$M_\ell C = C T_\ell + f_z(e_1 - e_2).$$

Here M_ℓ is the transition matrix, f_z the standard basis column supported at z, and e_i the color row; the displayed discrepancy is a rank-one outer product. Since S has one state in every cell, induction along each word w gives $\chi_S M_w C = \mathbf{1} + \sum_j \mu_j (e_1 - e_2) T_{w_j}$, where $\mu_j \geq 0$ counts trajectories at z immediately before the corresponding suffix w_j. A trajectory from X needs at least $t + 1$ steps to reach z; any entrance from P starts at y_1 after at least one step and then needs at least t more. Thus the first nonzero correction has position at least $t + 2$. In the prescribed cell graph, color 2 needs $t - 1$ letters to enter P at p_{p-1} and at least $p - r$ more to reach A. The only interchange is between adjacent last P-points and cannot shorten this descending path. A negative correction therefore reaches A only at length at least $(t + 2) + (t - 1 + p - r) = p + 2t + 1 - r = q$. Before then every protected coordinate has multiplicity at least one.

□

Lemma 6 (Attaining words)**.** *The words $a^{2t}bab^{p-r-1}$ and $a^{2t}ba^tb$ avoid A and compress S, respectively. Their lengths are q and L.*

Proof. After $a^{2t}b$, the missing P-point is p_{p-1}; after a further a it is p_{p-2}. This is the unique b^{p-r-1}-preimage of p_{r-1}. Hence $a^{2t}bab^{p-r-1}$ omits that protected point at length q. The $X \cup Y \cup \{z\}$ portion after $a^{2t}b$ is $\{x_2, \ldots, x_t, y_1, z\}$. After a^t it contains x_t and y_t, so $a^{2t}ba^tb$ compresses at length L. The existence of x_t in the preceding set uses $t \geq 2$.

□

Lemma 7 (Strong connectivity and synchronization)**.** *The automaton is strongly connected and synchronizing, and S is a reachable image.*

Proof. Put $B = P \cup Y$, $h = p + t$ and $E = B \cup \{z\}$. Letter b cycles B in the order $p_0, y_1, \ldots, y_t, p_{p-1}, \ldots, p_1$, and $Qb^t = E$. The a-cycle connects all other states to B, proving strong connectivity. On E, $d = a^{2t}b^h$ fixes P, cycles $y_1 \to y_t \to \cdots \to y_1$, and sends z to y_t. Thus $e = d^t$ fixes B and sends z to y_1. Since $Ba^{t+1} = S$, the word b^tea^{t+1} reaches S. The word a^{2t+1} fixes $X \cup Y \cup \{z\}$ and transposes the last two P-points, adjacent on the b-cycle. These words generate every permutation of B, with inverses realized by positive powers. Moreover $Ba = (B \setminus \{y_1\}) \cup \{z\}$, $Bab = (B \setminus \{y_2\}) \cup \{z\}$, and $Babe = B \setminus \{y_2\}$. Permutations route any pair in a current image into the kernel of abe; repeated compression synchronizes B. Prepending b^te synchronizes Q.

□

Proof of Theorem 3. Lemmas 4–6 give $\mathrm{ac}(S, A) = \min(q, L)$, and $L = 3m - 1$. Lemma 7 verifies the structural requirements. When $q \leq L$, equality in (1) follows. □

3.2 The remaining parameter range

Proposition 8 (Binary sharpness below the two-block threshold)**.** *Let $m \geq 4$ and $m+1 \leq q \leq 2m+1$. For every admissible r there is a binary strongly connected synchronizing automaton with a reachable $(n-m)$-image S and an r-set $A \subsetneq S$ such that $\mathrm{ac}(S, A) = q$.*

Proof. Appendix C supplies all three required constructions, including their reaching and reset words. The first cardinality face covers $q = m + 1$; the second covers $q = m + 2$. For $m + 3 \leq q \leq 2m + 1$, take $g = q - m - 1$ and $M = q - 1$ in Proposition 35. Then $g \geq 2$, $M \geq 2g$, $n = r + M + 1$, $s = r + g + 1 = n - m$, and $\mathrm{ac}(S, A) = M + 1 = q$. □

4 Reduction to finite configurations

Let P_h join distinct states if some word of length at most h merges them. Write $\tau(P_h)$ for its minimum vertex-cover size. A set survives compression through depth h exactly when it is independent in P_h. Thus

$$\exists T,\ |T| = n - m,\ c(T) > h \quad \Longleftrightarrow \quad \tau(P_h) \leq m. \tag{2}$$

This is the short-pair viewpoint on subset compression; see also the computational compression framework in [4]. Let $K_h = \{v \in Q : \deg_{P_h}(v) > 0\}$ be the nonisolated core and $J_h = Q \setminus K_h$ the isolated-state reservoir. If T is an independent $(n-m)$-set, its complement $C = Q \setminus T$ is a vertex cover of size m. As isolated vertices cover no edges, $C \cap K_h$ is also a vertex cover. Consequently

$$|(Q \setminus T) \cap J_h| \leq m - \tau(P_h). \tag{3}$$

In particular, when $\tau(P_h) = m$, every such T contains all of J_h:

$$J_h \subseteq T, \qquad Q \setminus T \subseteq K_h.$$

The cover number therefore controls how omitted states can lie outside the core. We use this observation together with the growth of short images.

Throughout this section, the automaton is strongly connected and synchronizing. For a word u, write $d(u) = n - |Qu|$.

Lemma 9 (Safety and union of actual short images)**.** *Suppose $|S| = n - m \geq 2$, $c(S) \geq L$, and $0 \leq d \leq L - 2$. Every Sw with $|w| \leq d$ has size $n - m$ and is independent in P_{L-1-d}. In a strongly connected automaton their union has at least* $\min(n, n - m + d)$ *states.*

Proof. Compression in the prefix w, or a merging suffix of length at most $L - 1 - d$, would compress S before length L. Let $U_i = \bigcup_{|w| \leq i} Sw$. If U_i is proper, strong connectivity gives an edge from U_i to its complement, whence $|U_{i+1}| \geq |U_i| + 1$. Starting at $|U_0| = n - m$ proves the union bound. □

Definition 10. Fix layers $P_h \subseteq P_H$, and let $\mathcal{D}$ be the complete family of maximum independent sets of P_H. A *row* is the characteristic vector of a member of $\mathcal{D}$; row rank means rank over $\mathbb{R}$. Let E consist of the isolated vertices of P_h omitted by at least one member of $\mathcal{D}$. The *distinguished set* is $W = K_h \cup E$, and the *reservoir* is $J = Q \setminus W$. Thus every row contains J. A *configuration* consists of the early graph, the later row family, and this partition of the states. A *letter signature* records the action on W, the reservoir entrance and exit counts, and the induced row images.

The principal reductions are stated below. Their proofs, which use independent-set bounds and the linear span of image vectors, appear in Appendix B. The codimension-three input is proved separately in Appendix A.

4.1 Image rank and finite support

The span of image vectors grows until compression. Applied to the kernel blocks of a word, this gives the following rank-transfer estimate.

Lemma 11 (Rank transfer)**.** *Let u have length a and rank $k \geq 2$ in a synchronizing automaton. Every k-set T satisfies*

$$c(T) \leq a + n - k + 1.$$

Lemma 12 (Iterated rank descent)**.** *For $m \geq 1$, $|S| = n - m \geq 2$, and a word u of length a and defect $d = d(u) \geq 1$,*

$$c(S) \leq \begin{cases} a, & d > m, \\ 2^{m-d}(a + d + 2) - 1, & 1 \leq d \leq m. \end{cases}$$

Lemma 13 (Rank of the image family)**.** *For a nonempty subset S of a strongly connected synchronizing automaton, the span V_i of the trajectory rows $\chi_S M_w$ for $|w| \leq i$ has dimension at least* $\min(i + 1, n)$*. Consequently, if these images are uncompressed members of a family $\mathcal{F}$, then* $\operatorname{rank} \mathcal{F} \geq \min(i + 1, n)$.

Lemma 14 (Bound on the distinguished set)**.** *Suppose $|S| = n - m \geq 2$, $c(S) \geq h + m + 1$, $h \geq 1$, and $1 \leq \ell \leq m - 1$. Let $G = P_h$, $H = P_{h+\ell}$, $t = \tau(G)$, $d = m - t$, and suppose $\alpha(H) = n - m$. For the complete maximum family $\mathcal{D}$ of H, put $I_0 = \bigcap \mathcal{D}$, $A_0 = Q \setminus \bigcup \mathcal{D}$, $a = |A_0|$, and $W = K_h \cup (Q \setminus I_0)$. Then $a \leq \ell$, $|W| \leq 2m + ad$; if $d \geq 1$ and $\ell \leq 2d$,*

$$|W| \leq 2m + \ell d - 1.$$

Each member of $\mathcal{D}$ omits at most d isolated vertices of G. In particular, bounds on $|W|$ for $(m, h, \ell, t) = (4, 7, 2, 3)$ and $(5, 9, 2, 4)$ are 10 and 11, respectively; the former already follows from $2m + ad \le 10$.

Lemma 15 (Degree, size, and rank bounds for the core)**.** *Under $|S| = n - m \ge 2$, $c(S) \ge L$, and $1 \le h \le L - m - 1$, put $t = \tau(P_h)$ and $B_m = \binom{m+2}{2}$. Then*

$$\Delta(P_h) \le m, \qquad |K_h| \le t(m - t + 2), \qquad |E(P_h)| \ge h.$$

If $0 \le b = B_m + h - L \le 3m - 2$, then $|E(P_h)| \le b$. When $t = m$, the maximum-independent-set row rank is at least $\min(L - h, n)$.

Lemma 16 (The cover-four branch with five missing states)**.** *Suppose $n \ge 16$, $|S| = n-5$, $c(S) \ge 15$, and $\tau(P_9) = 4$. Then $|K_9| \le 11$. If $|K_9| = 11$, the maximum-independent-set family of P_9 has exactly two members differing by one state; P_{10} retains exactly one of these members, which every letter maps into that same two-member family.*

Lemma 17 (Size of the later independent-set family)**.** *Let $G \subseteq H$ be graphs with $\alpha(G) = s + 1$ and $\alpha(H) = s$. Write $\mathcal{M}$ for the maximum independent sets of G, $\mathcal{B}$ for its inclusion-maximal independent sets of size s, and $\mathcal{D}$ for the maximum family of H. Then $|\mathcal{D}| \le |\mathcal{B}| + 2|\mathcal{M}|$. If Z is the union of the G-isolates omitted by members of $\mathcal{D}$, then $|Z| \le \max(2, |\mathcal{M}|)$.*

4.2 Compatible letter actions

The support bounds leave a finite family of configurations. To rule one out, we compare the actions of all letters on its independent sets and short pairs.

Lemma 18 (Constraints on letter signatures)**.** *Let $\mathcal{A} = (Q, \Sigma)$ be a strongly connected synchronizing n-state automaton. Fix integers $h \ge 1$, $H \ge h + 1$, and $L > H$. Suppose $|S| = n - m \ge 2$, $c(S) \ge L$, and $\alpha(P_H) = n - m$. Let $\mathcal{D}$ consist of all maximum independent sets of P_H, and define the distinguished set and common reservoir relative to $G = P_h \subseteq P_H$ and $\mathcal{D}$. The signatures induced by Σ satisfy the following conditions:*

1. *For every $D \in \mathcal{D}$ and $f \in \Sigma$, $|Df| = n - m$ and Df is independent in P_h. Thus Df has size $\alpha(P_h) - (m - \tau(P_h))$.*
2. *The pairs of P_h have merging distances at most h. The merging distances satisfy the one-step recurrence for Σ, with distance zero on a merged pair.*
3. *The union of the projected transitions of all letters on the distinguished states and the common reservoir is strongly connected.*
4. *If $H = h + 2$, then $|Df^2| = n - m$ and Df^2 is independent in P_h for every $D \in \mathcal{D}$ and $f \in \Sigma$. The two applications use the same action on the distinguished states.*

Moreover, $S \in \mathcal{D}$. The configuration therefore admits a selection of signatures with these properties and with the row and pair distances induced by $\mathcal{A}$.

4.3 The two compression thresholds

At the thresholds needed here, these reductions take the following form.

Lemma 19 (Seven-layer reduction)**.** *Under $n \geq 11$, $|S| = n-4$ and $c(S) \geq 12$, the branch $\tau(P_7) = 4$ is confined to nonisolated cores with $5 \leq |K_7| \leq 8$, degree at most four and $7 \leq |E(P_7)| \leq 10$. The maximum-independent-set family has row rank at least five.*

Lemma 20 (Nine-layer reduction)**.** *Under $n \geq 16$, $|S| = n - 5$ and $c(S) \geq 15$,*

$$\tau(P_9) \in \{4, 5\}.$$

The cover-five branch has nonisolated support between six and ten states, at least nine early edges, maximum degree at most five and maximum-family row rank at least six.

Lemma 21 (Auxiliary maximum source)**.** *Suppose $n \geq 16$, $|S| = n-5$, $c(S) \geq 15$, $\tau(P_9) = 4$ and the nonisolated support of P_9 contains eleven vertices. Then $\tau(P_{10}) = 4$. The maximum-independent-set family of P_9 consists of two sets R, T differing at one state. Exactly one member $R_0 \in \{R, T\}$ remains independent in P_{10}, and every letter maps this same R_0 into $\{R, T\}$.*

Lemma 22 (Core entrance bound)**.** *Let $G = P_h$ in a synchronizing automaton, with $h \geq 2$ and proper nonisolated support $V \subsetneq Q$. If a letter f sends an isolated state into $x \in V$, then*

$$\deg_G(x) \leq d(f) + F_G(f) \leq |E(G)| - h + 2,$$

where $d(f) = n - |Qf|$ and $F_G(f)$ counts distinct edges in G having a G-nonedge preimage under f.

Remark 23 (An additional entrance condition)**.** Let P_h be a proper short-pair layer of a synchronizing automaton, $h \geq 1$. Some letter f sends a pair outside P_h to an edge of P_h.

5 Computer-assisted verification

The reductions above associate to a putative counterexample a finite configuration: an early short-pair graph, a complete later row family on its distinguished set, and the common isolated reservoir. We test whether this configuration admits a collection of letter signatures with compatible pair and row distances. The alphabet is not fixed in advance.

5.1 The feasibility predicate

For a configuration $\mathcal{C}$, let $\Pi_0(\mathcal{C})$ denote the following finite feasibility problem. Its variables select signatures from the complete admissible pool and assign integer distances to the retained pairs and rows. Let $\mathcal{F}$ be the retained equal-size row family: the complete early maximum family in an equal-cover branch, and the complete later maximum family otherwise. Write H for its layer index, taking $H = h$ in the former case, and put $D = L - H$. For $R \in \mathcal{F}$, its exit distance is

$$e(R) = \min\{|w| : |Rw| < |R| \text{ or } Rw \notin \mathcal{F}\}.$$

An exited row is represented by a terminal object of distance zero. The slow-row condition is $\max_{R \in \mathcal{F}} e(R) \geq D$; the values of D are $5, 3, 6, 4$ for $(m, \tau(P_h)) = (4,4), (4,3), (5,5), (5,4)$, respectively. Indeed, the initial set cannot exit before depth $L - H$. For $\rho = \operatorname{rank} \mathcal{F}$ and $\mathcal{F}_j = \{R \in \mathcal{F} : e(R) > j\}$, the safe-tail conditions are

$$\operatorname{rank} \mathcal{F}_j \geq D - j \quad (0 \leq j < D), \qquad \operatorname{rank} \mathcal{F}_j \leq \rho - j \quad (\mathcal{F}_j \neq \varnothing, \rho < n).$$

For the lower bound, the images of a slow row through depth $D-j-1$ belong to $\mathcal{F}_j$, so Lemma 13 applies. For the upper bound, put $V_0 = \operatorname{span}\{\chi_R : R \in \mathcal{F}\}$ and recursively

$$V_{i+1} = V_0 \cap \bigcap_{f\in\Sigma} \{x : xM_f \in V_i\}.$$

Then $\chi_R \in V_j$ for $R \in \mathcal{F}_j$. A plateau makes the space invariant under all letters. If it contains a vector of nonzero coordinate sum, a reset word and paths to every state imply that it contains every coordinate vector, contradicting $\dim V_0 = \rho < n$. If $\mathcal{F}_j$ is nonempty, its characteristic vector has nonzero coordinate sum, so the first j inclusions are strict. This proves $\dim V_j \le \rho - j$. The exit distances are positive and at most $|\mathcal{F}|$, or at most ρ when $\rho < n$.

Pair variables record positive merging distances at most h for edges of P_h, with a merged pair terminal at zero. Each level $1, \dots, h$ occurs. If a selected signature sends a nonedge of P_h to a retained edge, that edge has distance h. An image pair outside P_h has distance greater than h, so its one-step upper inequality places no additional restriction on a retained pair.

1. Selected signatures preserve the cardinality and independence conditions on safe row images. The distance variables satisfy the prescribed early and later pair layers, the required slow-row condition, and the safe-tail rank bounds.

2. For every selected signature f and every retained nonterminal pair or row x (writing $d_R = e(R)$ for rows), the one-step inequality
$$d_x \le 1 + d_{xf}$$
holds, with distance zero at a terminal object. Every nonterminal object has a selected signature attaining equality.

3. The union of the distinguished-state and reservoir arrows of the selected signatures is strongly connected.

4. In branches with later layer P_{h+2}, a letter and its square satisfy the corresponding safe-row conditions, with consistent state identities in both applications. In the eleven-point early-support branch, one auxiliary source R_0 from Lemma 21 is chosen for the whole alphabet. The two possible choices are tested separately.

Other branch-specific power and defect bounds are those established in the reductions. The strong-connectivity condition concerns the selected alphabet, not each signature separately. With the reservoir contracted to a vertex $*$, it means that every distinguished vertex both reaches $*$ and is reachable from $*$ in the union of selected arrows. The weaker proper-core exit condition requires only the former paths. Some branches are already excluded by this weaker condition; such an exclusion also excludes Π_0.

The lower-cover tests also retain short-image union growth. Among the slow rows there must be an R for which, at every $0 \le j < D$,

$$\left| \bigcup_{|w|\le j} Rw \right| \ge \min(n, n-m+j).$$

Here words use the compatible signature pool and all these images remain in $\mathcal{F}$. The reservoir is common to the rows and cancels when the inequality is evaluated on their distinguished parts. A pool containing the actual alphabet must satisfy this condition.

An automaton realizing $\mathcal{C}$ supplies a solution of $\Pi_0(\mathcal{C})$: select its signatures and use its true shortest merging and exit distances. The shortest-distance recurrence gives condition (2); projection of state paths gives condition (3). Conditions (1) and (4) follow from the short-image and auxiliary-source lemmas. Thus infeasibility excludes a counterexample. Feasibility, conversely, need not produce an automaton, since the finite signatures retain only the specified constraints.

5.2 Enumeration of configurations

The support bounds make the graph generation finite. In the cover-five branch we generate all isomorphism classes on the allowed orders and apply the degree, edge, cover and row-rank bounds; disconnected graphs are included. For a lower-cover core, fix a minimum cover C. Its independent complement is specified by a multiset of nonempty neighborhoods in C, together with an arbitrary edge set inside C. Enumerating these data includes every eligible graph.

Equal-cover branches use the complete maximum-independent-set family of the early graph. In the lower-cover branches, the candidate later rows are the early graph's maximum independent sets with one point deleted and its inclusion-maximal near-maximum independent sets, including the permitted exceptional isolates. Write these candidates as $R_1, \dots, R_b$. For a family $\mathcal{F}$, let $U(\mathcal{F})$ consist of all pairs co-occurring in its rows, including diagonal pairs, and set

$$\mathrm{cl}(\mathcal{F}) = \{R_i : \text{ every pair in } R_i \text{ belongs to } U(\mathcal{F})\}.$$

The complete family of later maximum independent sets is closed under this operation. A family is rejected if its pair data also supports an early maximum set, contradicting the prescribed later independence number. The remaining rank, union and cardinality tests are supplied by the finite-support reductions.

Algorithm: closed row families in a lower-cover branch
Input: an ordered candidate list $R_1, \dots, R_b$.
Start with Visit($\mathrm{cl}(\varnothing), 1$).
Visit($\mathcal{F}, j$):
  If $\mathcal{F}$ passes all filters, emit its canonical representative.
  For $i = j, \dots, b$ with $R_i \notin \mathcal{F}$:
    Set $\mathcal{F}' = \mathrm{cl}(\mathcal{F} \cup \{R_i\})$.
    If some $R_t \in \mathcal{F}' \setminus \mathcal{F}$ has $t < i$, skip this branch.
    If the pair data supports an early maximum set, skip this branch.
    Otherwise call Visit($\mathcal{F}', i + 1$).

To see completeness, fix a target closed family and at each stage add its first missing row. Closure remains inside the target, so this path is not rejected. Conversely, the earlier-row rule determines a unique path to each closed family. Equivalent families are identified under core automorphisms and a consistent permutation of exceptional isolates.

For a fixed configuration, the signature enumeration runs over the admissible kernel partitions on distinguished states, the incoming images of reservoir states, and injections of the kernel blocks into available output labels. It records pair images, row images and reservoir cardinalities. This is a finite representation of every admissible one-letter action, rather than a restriction on alphabet size.

Proposition 24 (Finite verification)**.** *For each branch in Table 1, take all early graphs satisfying the structural reductions, all closed later row families on the bounded distinguished set, and each applicable auxiliary-source choice. Generate their complete signature pools and test* Π_0*, together*

with the branch's proper-core exit condition. The resulting exclusions are those in Table 1. The only configuration remaining in the largest branch has early core $K_{4,4}$ and the row family displayed in Section 6.2. It is excluded by Lemma 25.

m	Branch	Inputs	Infeasible	Remaining step
4	$\tau(P_7) = 4$	91 cores	—	proper-core exit
4	$\tau(P_7) = 3$	10,142	10,142	none
5	$\tau(P_9) = 5$	2,294 cores	2,294	none
5	$\tau(P_9) = 4$, $N = 11$	3,595	3,595	none
5	$\tau(P_9) = 4$, $5 \le N \le 10$	3,206,988	3,206,987	Lemma 25

Table 1: Finite verification after structural filtering; N is the early nonisolated support size. The first branch is resolved by its proper-core exit test after joint distance enumeration.

Remark (pair entrance). Remark 23 gives an additional condition: some selected letter sends a pair outside a proper layer into that layer. Adding this condition to Π_0 excludes the exceptional configuration as well, as verified in an independent selected-configuration check. The proof above uses the multipartite overlap lemma for this case. Every infeasibility conclusion for Π_0 remains valid after adding the entrance condition.

5.3 Verification methods

The computation compares two complete signature-generation orders, checks bounded graph and closed-family enumerations against literal enumeration, and compares the joint distance encoding with an independent bounded feasibility calculation. The largest calculation used CaDiCaL 3.0.1 on 24 workers and took about 6.4 hours. These comparisons test enumeration and encoding consistency; they do not replace the complete branch calculations.

The row ranks computed over $\mathbb{F}_{1009}$ equal rational ranks. A zero–one minor of order at most ten has absolute determinant below 522, by the bordered Hadamard bound $(k+1)^{(k+1)/2}/2^k$. At eleven named vertices the only larger minor has order eleven and all rows have weight six. Its determinant is divisible by six, by replacing one column by the sum of all columns, and has absolute value at most $1458 < 6 \cdot 1009$. Thus a nonzero minor cannot vanish modulo 1009, including in the subfamilies used for safe-tail rank tests.

No DRAT or LRAT certificates are supplied for the reported UNSAT calculations. Their verification therefore relies on the implementation, the solver results and the independent bounded comparisons described above. The available evidence is not a separately proof-checked SAT certificate collection.

Data and code availability

The implementation and reproduction instructions are available at the fixed repository revision cited in [8]. The accompanying `docs/RUNNING.md` identifies the source files, input partitions, solver version, commands and completion records, and separates full calculations from bounded validation runs. The cited source is revision `d951829`.

6 The cases of four and five missing states

We now apply the reductions to prove the upper bounds. The binary examples in Section 3 then give equality.

6.1 Four missing states

Let $n \geq 11$, $|S| = n - 4$, and suppose that $c(S) \geq 12$. The Frankl–Pin bound gives $C_2 \leq 6$, while Theorem 27 gives $C_3 \leq 9$. Together with (2), these imply

$$\tau(P_7) \in \{3, 4\}, \qquad \tau(P_9) = 4.$$

If $\tau(P_7) = 4$, the images of S under words of length at most four are maximum independent sets of P_7, and their union is Q. Lemma 19 bounds the core size, degree, edge count and row rank. This is the first branch of Table 1, which the finite verification excludes.

If $\tau(P_7) = 3$, use the complete maximum-independent-set family of P_9. Lemmas 14 and 17 bound its support and number of rows; Lemma 13 gives row rank at least three. At most two states are absent from its union. The resulting configurations form the second branch of Table 1. They are excluded by the joint distance, connectivity and square conditions. In both branches the support is proper because $n \geq 11$.

It follows that $c(S) \leq 11$. When $n \leq 10$, the bound $q \leq 9$ in (1) suffices. The matching examples are furnished by Proposition 8 for $5 \leq q \leq 9$ and Theorem 3 for $q \geq 10$.

6.2 Five missing states

Let $n \geq 16$, $|S| = n - 5$, and suppose that $c(S) \geq 15$. The codimension-three bound gives $\tau(P_9) \in \{4, 5\}$. In the cover-five case, short images of depth at most five are maximum independent sets whose union is Q. Lemma 20 therefore applies. Each letter has rank defect at most three: for defect four, Lemma 12 would give

$$c(S) \leq 2^{5-4}(1 + 4 + 2) - 1 = 13,$$

and larger defects give smaller bounds. The required slow row has exit distance at least six. In particular, Sf^j is independent in P_9 for every letter f and $0 \leq j \leq 5$. These are the cover-five configurations excluded in Table 1.

Suppose instead that $\tau(P_9) = 4$. The result just proved for four missing states implies $\alpha(P_{11}) = n - 5$. Let $\mathcal{D}$ be its complete maximum-independent-set family. Its row rank is at least four, at most two states are absent from its union, and its finite support has size at most eleven. Each row is either obtained by deleting one point from an early maximum independent set or is an inclusion-maximal near-maximum independent set.

For an eleven-point core, Lemma 21 supplies two early maximum sets R, T differing at one state and a single source $R_0 \in \{R, T\}$ that is independent in P_{10}. Every letter maps R_0 into $\{R, T\}$. The corresponding configurations are excluded by the eleven-point branch of the finite verification. For smaller cores, Lemma 22 and the joint tests leave only the configuration in Figure 2.

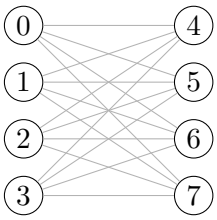
The following lemma excludes this last possibility.

Lemma 25 (Multipartite overlap exclusion)**.** *Let $h \geq 1$ and $t \geq 2$. Suppose a noncomplete P_h in a synchronizing automaton is a complete multipartite graph on $V = V_1 \sqcup \cdots \sqcup V_t$ and has isolated remaining states J. Assume each V_i admits a family $\mathcal{R}_i$ of nonempty subsets whose union is V_i, whose intersection graph is connected, and for which every $R \cup J$ with $R \in \mathcal{R}_i$ is independent in P_{h+1}. Then this configuration is impossible.*

Proof. A P_h nonedge cannot be merged in one letter. Fix a letter f. It is therefore injective on each part and on J, and Jf is disjoint from Vf. Since every $(R \cup J)f$ is P_h-independent, if $Jf \subseteq J$, then $Jf = J$ and Rf is contained in some single part. Intersecting R, R' have intersecting images, so the connected intersection graph forces all of $V_i f$ into one part. Every nonedge remains a nonedge.

Otherwise Jf meets some part V_j. Independence of $(R \cup J)f$ forces $Rf \subseteq V_j \cup J$ for every R, hence $Vf \subseteq V_j \cup J$ and $Jf \subseteq V_j \cup J$. Again every nonedge remains a nonedge. This holds for every letter. But a synchronizing automaton with noncomplete P_h must have some letter sending a nonedge to an edge of P_h: follow the shortest merging word for a pair at distance greater than h until the distance first drops from $h+1$ to h. Contradiction. □

In the exceptional configuration, the two intersecting triples in each part cover that part, and their extensions by J are independent in P_{11}. Lemma 25, applied with $h = 9$, rules it out. This proves $c(S) \leq 14$. For $n \leq 15$, the bound $q \leq 14$ suffices. The examples in Proposition 8 and Theorem 3 attain the upper bound for every remaining parameter value, completing the proof of Theorem 1.



$$R_1 = \{0,1,2\} \cup J$$
$$R_2 = \{0,1,3\} \cup J$$
$$R_3 = \{4,5,6\} \cup J$$
$$R_4 = \{4,5,7\} \cup J$$

Figure 2: The exceptional configuration: core $K_{4,4}$ and its four later rows. Every vertex of the isolated set J belongs to every row.

7 Conjecture and open problems

The binary lower-bound constructions give $\mathcal{F}_{\text{bin}}(n, n-m, r) \geq \min(q, 3m-1)$ for every $m \geq 4$, and Theorem 1 establishes equality for $m = 4, 5$.

Conjecture 26 (Linear defect formula)**.** *For every $m \geq 4$ and all admissible n, r,*

$$\mathcal{F}_{\text{bin}}(n, n-m, r) = \mathcal{F}(n, n-m, r) = \min(n-r, 3m-1).$$

The remaining parameter region for exact avoid-or-compress extrema is $m \geq 6$ and $q \geq 3m$.

A Compression with three missing states

Theorem 27 (Compression with three missing states)**.** *In a strongly connected synchronizing automaton with $n \geq 6$, every $(n-3)$-set has a compressing word of length at most nine.*

The proof proceeds by classifying the first three merging pairs. We first establish the common reductions, then exclude the nine possible types. Throughout this section, a string such as 012 denotes the set $\{0,1,2\}$, and the holes of an image are its complement in Q. For a family of holes, containment of R_i means containment of the complements of the images in R_i. A core row lists the images of the core states in increasing order; an asterisk denotes a free entry. We use $[E]$ for the indicator of a condition E.

A.1 Common reductions

Assume $c(S) = 10$, which is the only remaining case under the Frankl–Pin bound. The refinement of [4] gives $c(S) \le 10 - p(P_h) + h$, where $p(P_h)$ is the maximum length of an $(n-3)$-set Frankl–Pin sequence using pairs from P_h. Any $q \le 4$ distinct pairs admit a sequence of length q: for its ith pair choose three holes outside that pair, including an endpoint outside it of each later pair, and fill the remaining holes arbitrarily. There are at most three required endpoints and $n - 2 \ge 4$ available states. The pair is in its own subset and absent from every earlier subset. Consequently $|P_h| \le h$ for $h = 1, 2, 3$. Pair layers grow strictly until complete, so $|P_h| = h$ at these three depths. In particular every letter is a permutation or has defect one with the common kernel 01.

Write $R_i = \{Sw : |w| \le i\}$ and $V_i = \operatorname{span}\{\chi_S M_w : |w| \le i\}$. Until compression $R_i \subsetneq R_{i+1}$: a plateau is forward invariant and contradicts synchronization. Also $\dim V_i \ge \min(i+1, n)$. Indeed a plateau in the vector spaces is invariant under all letters; a reset word followed by a path to any state yields all rows $|S|e_q$. Before compression these trajectory rows equal the characteristic rows of the actual images. Every image through depth six is independent in P_3. If $\mathcal{H}$ is its three-hole cover family and tests have maximum length L, the domain preserving $\mathcal{H}$ under these tests contains R_{6-L}. Such a domain therefore has at least $7 - L$ rows. Exiting it permits compression with at most $L + 3$ additional letters. For a permutation f, the hole action is $H \mapsto Hf$. For a defect-one letter with missing output e, it is $H \mapsto \{e\} \cup Hf$ when $01 \subseteq H$, and $H \mapsto \{e\} \cup (H \setminus \{0, 1\})f$ otherwise, for the safe covers under discussion. These formulas determine the hole families used below.

Rank and invariant-set obstructions. For the low-rank cases the following table lists the *occupied core sets*; a common true outside set J is adjoined to every row.

case	occupied core rows	contains
triangle C; type 4 closed	$24, 23, 14, 13, 04, 03$	R_5
type 4 mixing	$145, 135, 045, 035, 124, 123$	R_5
type 7 $D_{6,\mathrm{cl}}$	$14, 04, 13, 03, 12, 02$	R_5
type 7 $D_{6,T}$	$135, 035, 125, 025, 134, 034$	R_5
type 7 $D_{6,A}$	$145, 045, 125, 025, 134, 034$	R_5
type 7 C_A	$34, 13, 03, 12, 02$	R_4
type 7 C_V	$14, 13, 03, 12, 02$	R_4
type 7 M_w	$135, 035, 125, 025, 134$	R_4

In the first five rows, characteristic rows X_i in the displayed order satisfy $X_1 - X_2 = X_3 - X_4 = X_5 - X_6$; these are two independent relations. For C_A, C_V, $X_2 - X_3 = X_4 - X_5$; for M_w, $X_1 - X_2 = X_3 - X_4$. Every rank is at most four. All coefficients sum to zero, so adjoining J preserves the relations. This contradicts $\dim V_4 \ge 5$.

The following occupied-core families each form a five-cycle on five states:

type 2 mixing B / closed T	$14, 04, 23, 12, 03$
type 3 mixing T	$24, 04, 13, 03, 12$
type 6 closed II	$34, 14, 12, 03, 02$
type 7 C_T	$34, 14, 03, 12, 02$

Each contains R_4, with the common set J adjoined. Since every letter has defect at most one, $\dim(V_3 D_f) \ge 3$. If a point of J entered a core point z, every row of $V_3 D_f$ would be spanned by domain rows containing z; there are only two, a contradiction. Thus every letter preserves the nonempty proper J (nonempty because $n \ge 6$), contradicting strong connectivity. We will refer to these two obstructions in the case analysis.

Write $E_1 = \{0, 1\}$. Up to simultaneous state relabeling the ordered three pairs, for $n \geq 6$, have precisely these nine types:

Type	E_2	E_3	Obstruction
0	02	12	Four-image families; rank at most four
1	02	03	State 0 has no incoming path
2	02	13	Small families; five-cycle; short exit
3	02	23	Small families; five-cycle; short exit
4	02	34	Six-image families of rank at most four
5	23	02	Incompatible inverse pairs
6	23	04	Two-step bound; five-cycle
7	23	24	Two-step bounds; rank; five-cycle
8	23	45	Row space of dimension at most four

To see that the list is complete, note that E_2 shares one endpoint with E_1 or is disjoint. In the first case normalize it to $\{0, 2\}$; the ordered pair stabilizer fixes 0,1,2, and E_3 can be the remaining old edge, an edge from each of those three vertices to one new vertex, or an edge on two new vertices. In the disjoint case normalize E_2 to $\{2, 3\}$. Its stabilizer independently swaps the endpoints of the first two pairs. Thus E_3 joins the pairs, joins either pair to a new point, or uses two new points. These give the five and four types above.

A.2 Type 0: a triangle

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 02\}, \quad P_3 = \{01, 02, 12\}.$$

Throughout $C = \{0, 1, 2\}$. A hole set covers P_3 exactly when it is C or is $\{i, j, x\}$, with $\{i, j\}$ a core pair and x outside C. Every image reached within six letters must have such holes, else a P_3 word compresses the initial set within nine letters.

Letter structure The short-pair reduction forces every letter to be a permutation or a rank-$(n-1)$ map with common kernel 01. Inverse pairs of 01 lie in P_2; inverse pairs of 02 lie in P_3. A permutation therefore preserves C. A singular letter has missing output 0 or 1. If it omits 1, inverse pairs of 02 force it to map C onto $\{0, 2\}$. If it omits 0, write its kernel image $c = f(0) = f(1)$ and its other core image $t = f(2)$, with $c \neq t$ and c,$t \neq 0$.

Some permutation g satisfies $g^{-1}(01)$=02, since singular letters have no two-point preimages of 01. In particular $g(02) = 01$ and g preserves C and its complement. Strong connectivity requires a singular letter that does not preserve C: otherwise every letter preserves this proper set. Such a letter omits 0. Its possibilities are:

- A: c in $\{1, 2\}$, t outside C;
- B_1: c outside C, $t = 1$;
- B_2: c outside C, $t = 2$;
- C: c,t outside C.

Case A: exclusion For Case A, the domain D_f of P_3-covering holes whose f-images also cover P_3 consists of C, $\{0,2,v\}$, $\{1,2,v\}$, and all $\{0,1,x\}$, where v is the unique external preimage of the other point in $\{1,2\}$. Its intersection with $g^{-1}(D_f)$ has at most four elements. Because g sends 01 to a different core pair, at most one hole $01x$ survives its test, namely $x = g^{-1}(v)$; the other three holes contribute at most three. All images reached within four letters must be in that intersection, contradicting strict image growth.

Case B_1: exclusion Let v be the unique external preimage of 2. Then

$$D_f = \{C, 01v\} \cup \{02x, 12x : x \notin C\}.$$

Define $D = D_f \cap g^{-1}(D_f) \cap f^{-1}(D_f)$. All images reached within four letters lie in D: appending f, gf or ff must still cover P_3 within six letters.

The core pair 02 has at most one external choice in D, because g sends 02 to 01. For pair 12, membership in $f^{-1}(D_f)$ forces $x = v$ or $f(x) = v$, so there are at most two external choices. The pair 01 has at most its single choice v. The hole C belongs to $f^{-1}(D_f)$ only if $c = v$. If $c = v$, there is no external preimage of v (it is the kernel image), so the pair 12 has only its choice v. Thus $|D| \leq 4$ in both cases $c = v$ and $c \neq v$. Strict image growth within four letters gives a contradiction.

Case B_2: reduction to two four-image domains Now $f(2) = 2$. Let $v = f^{-1}(1)$ outside C and let $u = g^{-1}(v)$, also outside C. Again D_f consists of C, $\{0,1,v\}$, and all $\{0,2,x\}$,$\{1,2,x\}$. Put

$$D = D_f \cap g^{-1}(D_f) \cap f^{-1}(D_f) \cap (fg)^{-1}(D_f).$$

The tests correspond to f,gf,ff,fgf, all of length at most three. Every image reached within three letters therefore lies in D.

The four tests give the following restrictions:

- C is in D only if $c = u$ (test fgf);
- $\{0,1,v\}$ is in D only if $c = v$ (test ff);
- core pair 02 has at most external choice u (test gf); it further requires $u = v$ or $f(u) = u$ (test fgf);
- core pair 12 has at most external choices v and $w = f^{-1}(u)$, where w is included only if it exists outside C (test fgf).

If $c = u$, w does not exist outside C, since u is then the kernel image. If $c \neq u$, C is absent. These restrictions give $|D| \leq 4$. Strict growth forces equality and all four domain images to be reached within three letters. Equality has only two possibilities:

1. $c = u = v$. Then D is $\{012, 01v, 02v, 12v\}$.
2. $c = v \neq u$ and $f(u) = u$. Then D is $\{01v, 02u, 12v, 12u\}$.

Every letter must send every image in D to a P_3-covering image: each was reached within three letters. In particular the letter assumptions A and B_1 are excluded by the arguments above.

For the equal-parameter B_2 domain, the occupied core rows are the four singletons of $C \cup \{v\}$, with a common J. The four rows are independent and contain R_3. If J enters a core point under f,

every row of V_2D_f is proportional to the same occupied-core singleton row. But $\dim(V_2D_f) \geq 2$, a contradiction. Hence J is invariant, contradicting strong connectivity.

For the other B_2 domain, relabel $v = 3, u = 4$ and write its occupied rows $X_1 = J \cup 24$, $X_2 = J \cup 13$, $X_3 = J \cup 04$, $X_4 = J \cup 03$. These four rows are independent. The sole kernel direction of a singular letter is $k = e_0 - e_1 = X_4 - X_2$. A three-row subspace contains k exactly when its rows include X_2, X_4. If some letter f sends a point of J to a core point z, the output space V_2D_f is spanned by at most two domain rows containing z. Thus f is singular, $\dim V_2 = 3$, and R_2 consists of X_2, X_4 and one further row. At least two possible output rows must contain z. Permutations and missing-1 letters preserve 012, hence J can enter only 3 or 4 under those letters. Missing-0 letters have only one available row at 1 or 2; their rows without 0 are $24, 13$, disjoint, so every entry is impossible. For a missing-1 letter entry at 3 permits only 03. The only remaining possibility is missing-1 and $z = 4$, with outputs $24, 04$. Its kernel image is in $\{0, 2\}$, hence $X_2f = X_4f$. If the third row is X_3, it also contains 0, so all three outputs coincide. If it is X_1, then $X_1 \cup X_4$ excludes 1 and f is injective there, giving $|X_1f \cap X_4f| = |J|$. Equal output rows instead have intersection size $|J| + 2$, and distinct rows $24, 04$ have intersection size $|J| + 1$. Both alternatives contradict injectivity. Thus J has no entry into the core and is invariant, again impossible.

Case C: a six-image domain Let v_1,v_2 be the distinct external preimages of 1,2. The domain D_f is exactly the six holes formed by one core pair and one of v_1,v_2; C itself is not in D_f. Every image reached within five letters lies in D_f, because its f-image reached within six letters must cover P_3. Strict growth gives all six domain images by depth five. Every letter sends every domain image to a P_3-covering image.

The six occupied rows are the triangle-C low-rank family listed above, so the trajectory-rank contradiction excludes this case.

A.3 Type 1: a star

Suppose $P_1 = \{01\}$, $P_2 = \{01, 02\}$, $P_3 = \{01, 02, 03\}$. For every letter f, preimages of 01 are in P_2, and preimages of 02 are in P_3.

If f is a permutation, $f^{-1}(01)$ is 01 or 02, and $f^{-1}(02)$ is a different edge of the three-edge star. Their intersection is $\{0\}$. Since 01 and 02 also intersect at $\{0\}$, bijectivity gives $f^{-1}(0)$=0. In particular no other state maps to 0 under f.

If f is singular, let d be its unique missing output and c the common image of 0,1. If d is not in 01 and c is in 01, the preimages of 01 are the two distinct pairs $\{0, z\}$,$\{1, z\}$, for z outside $\{0, 1\}$; both cannot lie in P_2. If d and c are both outside 01, its unique preimage pair is disjoint from $\{0, 1\}$, again impossible in P_2. Hence d is 0 or 1. If $d = 1$, examine 02. When c is in 02, its two preimages $\{0, z\}$,$\{1, z\}$ cannot both belong to P_3. When c is outside 02, its unique preimage pair is disjoint from $\{0, 1\}$ and cannot belong to P_3. Thus $d = 0$. N_o state therefore maps to 0 under a singular letter.

Every letter either fixes 0 with no other incoming edge, or omits 0 entirely. State 0 is unreachable from every other state, contradicting strong connectivity.

A.4 Type 2: a path with opposite terminal pairs

We have $P_1 = \{01\}, P_2 = \{01, 02\}, P_3 = \{01, 02, 13\}$. Set $C = \{0, 1, 2, 3\}$.

The three-hole covers of P_3 are the four triples in C, together with $01x, 03x, 12x$ for $x \notin C$. Denote this family by H.

Compatible letters Every singular letter omits 0. Indeed its missing output must be 0 or 1, because otherwise inverse pairs of 01 either are disjoint from 01 or contain both $0z$ and $1z$, incompatible with P_2. If it omitted 1, inverse pairs of 02 would similarly either be disjoint from 01 or contain both $0z$ and $1z$. Neither possibility belongs to P_3. Write such a letter f as $f(0) = f(1) = c$, $f(2) = t$, $f(3) = z$, where c,t,z are distinct and nonzero.

Permutations fall into three classes, by their specified core rows:

- A: (0,1,2,*) on C;
- B: (0,2,1,*) on C;
- $T : (1, 0, *, 2)$.

The free image in B is 3 or outside C; that in T is $T_2 = 3$ or outside C. Call these closed and mixing respectively. There is a B permutation: the new inverse pair 02 of 01 must arise from a permutation, and class T has inverse pair 01 instead. There is also a T permutation: only it has inverse pair 13 of 02. Singular letters omit 0, so create neither of those new inverse pairs. The classification follows directly by intersecting the inverse pairs of 01 and 02; their common inverse endpoint is the preimage of 0.

Permutation mixing Direct hole substitution gives the following domain bounds. Juxtaposition of letters is right action, so TB means first T, then B.

Letters	Tests W	L	Maximum	Core holes and outside restrictions
B mixing, T mixing	B, T	1	5	012,023; at most one outside choice for each of 01,03,12
B mixing, T closed	B, TB	2	5	023,123; $01v$,$03v$,$12u$, where $v = B^{-1}(3)$, $u = T^{-1}(v)$
B closed, T mixing	B, T, TT	2	4	At most three core holes and one outside hole, or two core holes and two outside holes
B,T closed, A mixing	A, B, TA	2	4	012,013; at most $03v$ and $12T^{-1}(v)$, $v = A^{-1}(3)$

In the first row $v = B^{-1}(3)$ fixes the outside choice for 01 and 03, and $T^{-1}(3)$ fixes it for 12. The bound five contradicts $|R_5| \geq 6$.

In the second row, membership of $01v$ further requires $T(v) = v$. Equality five therefore forces $u = v$, and its domain is exactly

$$D = \{023, 123, 01v, 03v, 12v\}, \qquad v \notin C, \quad T(v) = v.$$

If its size is smaller than five, $|R_4| \geq 5$ gives a contradiction. The equality case is treated by the five-cycle argument above.

For the third row put $a = T_2$ outside C and $v = T^{-1}(3)$. If $v = a$, the possible core holes are 012,013,023 and the only outside hole is $12v$: $T^{-2}(3) = 2$ is in C. If $v \neq a$, only 012,013 remain among core holes, and at most $12v$,$03T^{-2}(3)$ remain outside. Both give the stated four. In the last row A preserves 012 and sends 3 outside. The B test deletes all $01x$; the A and TA tests restrict the other two types to the listed unique preimages. These two rows contradict $|R_4| \geq 5$.

The equality domain has the type-2 five-cycle occupied rows listed above, so the common-entry argument excludes it. Thus all permutations preserve C.

Singular mixing once permutations preserve C Choose a closed B and T. Strong connectivity requires a singular f that does not preserve C. Retain $f(0) = f(1) = c$, $f(2) = t$, $f(3) = z$. Say t or z is good when it belongs to $\{1, 3\}$. Direct substitution gives:

$Goodimagesamongt, z$	Tests	L	Domain bound
neither	B, f	1	4
exactly one	B, f, Tf	2	4
both	B, fB, TfB	3	4

For the first row B excludes $01x$. The holes 023,123 fail f. If c is 1 or 3, the two other core holes may remain, but only the other member of $\{1, 3\}$ has an outside preimage; each of $03x$ and $12x$ has at most that one choice. If c is neither, no core hole remains and each outside type has at most two choices. Both counts are at most four.

In the second row T swaps 012 with 013, so the tests f,Tf retain either both of these holes or neither. The holes 023,123 are similarly paired. Each outside type has at most the unique outside preimage of the other member of $\{1, 3\}$. If all four core holes remain, c must be that other member, so this outside preimage does not exist. Thus again the count is at most four. The first two rows contradict respectively $|R_5| \geq 6$ and $|R_4| \geq 5$.

In the last row $\{t, z\} = \{1, 3\}$; mixing forces c outside C. Let $v = f^{-1}(2)$, $u = T^{-1}(v)$, both outside C. The domain has exactly four holes:

$$D = \begin{cases} \{023, 123, 12v, 03u\}, & t = 1,\ z = 3, \\ \{023, 123, 03v, 12u\}, & t = 3,\ z = 1. \end{cases}$$

Both 023 and 123 have f-image holes 013, which is outside D, so two distinct vertices have exit distance one. Every vertex must exit under synchronization. In a four-vertex graph with two distinct distance-one vertices, maximum exit distance is at most three: a shortest distance-four path would use distinct vertices of all four levels, leaving only one vertex at level one. The initial vertex is in D. Its exit takes at most three letters; a failed defining test takes at most three more, and a P_3 merge at most three more. Thus compression has length at most nine. This excludes the last singular mixing possibility and contradicts strong connectivity.

A.5 Type 3: a path with adjacent initial pairs

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 02\}, \quad P_3 = \{01, 02, 23\}.$$

Let $C = \{0, 1, 2, 3\}$. Three-hole sets covering P_3 form the family H consisting of the four triples in C and $02x$,$03x$,$12x$ for x outside C.

Compatible letters Every letter is a permutation or a rank-$(n-1)$ map with common kernel 01. Its inverse pairs of 01 belong to P_2, and those of 02 belong to P_3. The permutation templates are:

$$A : (0, 1, 2, *), \qquad B : (0, 2, 1, *), \qquad T : (1, *, 0, 2).$$

A,B are closed when their last core image is 3; T is closed when $T_1 = 3$. Otherwise that image is outside C. Singular letters either omit 0, with $f(0) = f(1) = c$, $f(2) = t$, $f(3) = z$ distinct and nonzero, or omit 1 with one of

$$D_0 : (c, c, 0, 2), \qquad D_2 : (c, c, 2, 0), \qquad c \geq 3.$$

These last letters are closed when $c = 3$, otherwise mixing. To check completeness, a singular missing output is 0 or 1 by the inverse-pair 01 test. If it omits 1, its kernel image cannot belong to 02: that would create both $0x$ and $1x$ as inverse pairs, which cannot both lie in P_3. The inverse pair of 02 must therefore be the disjoint pair 23, giving the two displayed orientations. The permutation templates follow by intersecting the inverse pairs of 01 and 02. Growth from P_1 to P_2 requires B or T; growth from P_2 to P_3 requires T or D_0 or D_2.

A mixing T permutation Put $a = T_1$ outside C and $v = T^{-1}(3)$. Test T and TT. If $v \neq a$, their common domain is contained in $\{02v, 12v, 03v, 03T^{-1}(v)\}$, of size at most four, contradicting $|R_4| \geq 5$. If $v = a$, the domain is exactly

$$D = \{013, 123, 02v, 12v, 03v\}.$$

The occupied core rows are the type-3 five-cycle family listed above. The common-entry argument excludes mixing T.

A closed T permutation Now T on C is (1,3,0,2). Its outside action is a permutation. For an additional letter g, let $D_2(T, g)$ be the domain preserving H under all words of length at most two on T,g. Direct substitution gives:

$Mixing g$	Domain is contained in	$Definition of v, u$
A	$123, 03v, 12u$	$v = g^{-1}(3), u = T^{-1}(v)$
B	$012, 03v, 12u$	*same*
D_0	$013, 03u, 12v$	*same*
D_2	$023, 123, 12u$	*same*

For A, the tests g,gT,Tg remove respectively core 013,023,012; for B they remove 023,013,123. Their outside constraints require $gx = 3$ for $03x$ and $g(Tx) = 3$ for $12x$. In the D_0 case gg removes 023,123 and g removes 012; its 013 survives only if $g(g(0)) = 3$, equivalently $g(0) = v$, while the

outside restrictions give the displayed unique preimages. In D_2, g removes 013, Tg removes 012, and gg removes every possible $03x$. These observations verify all four rows. Every domain has size at most three, contradicting $|R_4| \geq 5$. Thus every permutation and every missing-1 letter preserves C. Strong connectivity requires a mixing missing-0 letter f with core images c,t,z.

Call a core output good if it is 2 or 3. If at most one of c,t,z is good, $D_2(T, f)$ has at most four vertices. With none good there are no core holes, and each outside type $03x$,$12x$ has at most the two outside preimages of 2,3. With only c good, there is at most core 012 and one outside choice for each type. With only t good, core holes are among 023,123, with at most $03v$,$12T^{-1}(v)$; with only z good they are among 013,123, with at most $12v$,$03T^{-1}(v)$. Here v is the outside preimage of the other good output. These counts are at most four.

If c and one of t,z are the two good outputs, the remaining core image is outside C (otherwise f is not mixing). All outside holes are removed by f and Tf. At least one core hole is removed by fT: if $c = 2$ it is 013 when z is outside, or 012 when t is outside; if $c = 3$ the holes 023,123 are removed. The domain then has size at most three.

It remains that c is outside and $\{t, z\} = \{2, 3\}$. Set $v = f^{-1}(1)$ and $u = T^{-1}(v)$, both outside C. Compute $D_3(T, f)$, the domain preserving H under all words of length at most three:

$$D_3 = \begin{cases} \{013, 123, 03u\}, & t = 2,\ z = 3, \\ \{023, 123, 03v\} \cup \{12u : f(u) = v\}, & t = 3,\ z = 2. \end{cases}$$

The two-step families before the last tests are

$$\begin{cases} \{013, 023, 123, 12v\} \cup \{03x : x \notin C\}, & t = 2,\ z = 3, \\ \{012, 023, 123, 03v\} \cup \{12x : x \notin C\}, & t = 3,\ z = 2. \end{cases}$$

Requiring their T- and f-successors to remain in those domains gives the displayed D_3 sets directly. Their sizes must be at least four because $|R_3| \geq 4$. Only $t = 3$,$z = 2$ and $f(u) = v$ survive. Since $f(v) = 1$, $u \neq v$. Relabel $u = 4$,$v = 5$ and order the remaining domain as

$$D = \{023, 123, 124, 035\}.$$

Here $T = (1, 3, 0, 2, 5, \ldots)$ and $f = (c, c, 3, 2, 5, 1, \ldots)$, with distinct unspecified outputs and $c \notin \{0, 1, 2, 3, 5\}$. For the holes $023, 123, 124, 035$, respectively, apply T, TT, ff, f. Each yields holes 012, outside the domain, without intermediate compression. For example $124f = 035$ and $123T = 023$. Thus every domain row exits within two letters. A defining test takes at most three letters, and merging a pair in P_3 takes at most three more. Thus $c(S) \leq 8$, a contradiction.

No T permutation There must now be a B permutation and a D_0 or D_2 letter. A closed D_0 is impossible even alone: its common domain for D_0,D_0D_0 is precisely the four core holes, fewer than $|R_4| \geq 5$. A mixing D_0 together with B also has a two-step domain of size at most four. To see this write $v = D_0^{-1}(3)$. The D_0 and D_0D_0 tests leave at most core 013, only if $D_0(v) = 3$ with $v = D_0(0)$, and outside holes $12v$,$02v$,$03x$ with $x = v$ or $x = D_0^{-1}(v)$. If B is closed it removes $02v$; if core 013 is present, $D_0^{-1}(v)$ is not outside, giving at most three vertices. If B is mixing, its test forces $x = B^{-1}(3)$ for $03x$ and for $02x$. There are then at most one core hole and three outside holes. Both counts are at most four. Thus no D_0 letter exists, and a D_2 letter is necessary.

If the alphabet consisted solely of A,B permutations and D_2 letters, the family of holes $12x$ ($x \neq 1$,2, including $x = 0$,3) would be invariant. A,B preserve the pair 12. A D_2 letter sends 120 to $12c$ and any other $12x$ to $12D_2(x)$; these are again three distinct holes. This invariant family of uncompressed $(n - 3)$-images contradicts synchronization. There is therefore a missing-0 letter f.

Choose one $B = b$, one $D_2 = d$ and such an f. Let β be 1 when b is mixing, ε be 1 when d is mixing. In their common H-domain, $02x$ has at most the single outside choice $b^{-1}(3)$, available only when $\beta = 1$; $03x$ has at most the outside choice $d^{-1}(3)$, available only when $\varepsilon = 1$. Core 023 is absent if $\beta = 1$, and core 013 is absent if $\varepsilon = 1$. Let $c = f(0) = f(1)$, $t = f(2)$, $z = f(3)$.

First suppose t is neither 2 nor 3. If neither c nor z is good, all core holes fail and each outside type $02x$,$12x$ requires fx in $\{2, 3\}$; there are at most two $12x$ and at most one of each other type, totaling four. If only c is good, core holes are among 012,013, and all outside types require the unique outside preimage v of the other good output. There are at most three outside holes; if all three are present, both b and d are mixing and core 013 is absent, again totaling at most four. If only z is good, core holes are among 013,023,123; the possible $02x$,$03x$ are offset by the respective absence of 023,013, leaving at most three core/outside-extra holes plus the one possible $12x$. If both c,z are good, $02x$ and $12x$ are impossible since neither good output has an outside preimage. The possible $03x$ is offset by the absence of core 013. In each case the one-step domain has size at most four.

Now suppose $t = 2$ or $t = 3$ and use all tests of length at most two on b,d,f. Define a two-element output-target set V by

$$V = \begin{cases} \{1, b^{-1}(3)\}, & t = 2, \\ \{2, w\}, & t = 3, \end{cases} \qquad w = \begin{cases} 1, & d \text{ closed}, \\ d^{-1}(3), & d \text{ mixing}. \end{cases}$$

The test fb in the first case, or fd in the second, forces fx in V for every allowed outside $12x$. Put $a = [c \in V]$, $h = [z \in V]$. The kernel output c and core output z have no outside preimages, so at most 2-a-h outside $12x$ survive. The same test requires c in V for core 012, and z in V for core 023,123. Hence the core count is at most

$$a + h(2 - \beta) + (1 - \varepsilon).$$

The last term bounds core 013, and the β term removes 023 under mixing B. The other two outside types contribute at most $\beta + \varepsilon$. The total domain size is therefore at most

$$\begin{aligned} a + h(2 - \beta) + (1 - \varepsilon) + (2 - a - h) + \beta + \varepsilon \\ = 3 + h(1 - \beta) + \beta \leq 4. \end{aligned}$$

This also covers the common-kernel issue in fd: when d is closed, preimages of output 3 are 0,1, but f omits 0, leaving only target 1. When d is mixing the preimage is the unique outside state $d^{-1}(3)$. All targets in V are distinct and neither equals t.

Thus the common two-step domain on b,d,f has at most four images, contradicting $|R_4| \geq 5$. This completes the case without a T permutation.

A.6 Type 4: an adjacent pair and a disjoint edge

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 02\}, \quad P_3 = \{01, 02, 34\}.$$

Write $C = \{0, 1, 2, 3, 4\}$ and $G = \{3, 4\}$. The three-hole sets covering P_3 form H: the seven core holes

$$013, 014, 023, 024, 034, 123, 124,$$

and $03x$,$04x$ for x outside C. Every image reached within six letters must have holes in H.

All compatible letters Every letter is a permutation or a rank-$(n-1)$ transformation with common kernel 01. Inverse pairs of 01 belong to P_2 and inverse pairs of 02 belong to P_3. Permutations have core prefix A=(0,1,2) or B=(0,2,1); they fix 0 and preserve the pair 12. An inverse pair 34 of 02 cannot occur in a permutation: its inverse pair of 01 is 01 or 02 and would be disjoint from 34, whereas the target pairs intersect at 0.

A singular letter omits 0 or 1, by the inverse-pair 01 test. If it omits 1, its kernel image cannot be 0 or 2: that would produce both $0x$ and $1x$ as inverse pairs of 02, incompatible with P_3. Its unique inverse pair of 02 must therefore be 34. It has one of the core rows

$$(c, c, t, 0, 2), \quad (c, c, t, 2, 0), \qquad c, t \geq 3, \quad c \neq t.$$

A singular letter omitting 0 has unrestricted distinct nonzero images on the common kernel and the other core states. Some missing-1 letter is necessary, because only such a letter can add the new pair 34 to P_3. Call it d. Swapping 3 and 4 if needed, write d=(c,c,t,0,2) on C. This relabeling preserves the whole early-pair type.

The domain of d has six images Let D be the holes in H whose d-images also have holes in H. The hole 034 always has d-image holes 012, so is absent. Core holes 013,014 survive exactly when c is in G; holes 023,024,123,124 survive exactly when t is in G. An outside $03x$ or $04x$ survives exactly when dx is in G. Among the two outputs in G, every one not occupied by c or t has one outside preimage. Thus, writing $a = [c \in G]$, $b = [t \in G]$,

$$|D| = 2a + 4b + 2(2 - a - b) = 4 + 2b.$$

For example, 013 has holes $\{1, c, 0\}$, 014 has $\{1, c, 2\}$; 023 and 123 have $\{1, t, 0\}$, 024 and 124 have $\{1, t, 2\}$; outside holes have $\{1, 0, dx\}$ or $\{1, 2, dx\}$. In each case the remaining condition is to meet G.

All images reached within five letters belong to D; hence $|D| \geq 6$. We must have t in G and $|D| = 6$, so every D-image is reached within five letters. In particular every letter sends every D-image to an H-image, because the resulting depth is at most six.

There are exactly two domain shapes:

- c in G: c,t are the two members of G and $D_{\text{closed}} = \{013, 014, 023, 024, 123, 124\}$;
- c outside C: let v be the unique outside preimage under d of the other member of G. Then $D_{\text{mixing}} = 023, 024, 123, 124, 03v, 04v$.

Both occupied row families are the type-4 low-rank families listed above; their rank is at most four, contradicting the trajectory rank.

A.7 Type 5: a crossing edge

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 23\}, \quad P_3 = \{01, 23, 02\}.$$

where P_h denotes the unordered distinct-state pairs merging in at most h letters. N_o strong-connectivity, synchronization, alphabet-size or reachable-image assumption is needed. The labels require $n \geq 4$.

The single pair in P_1 forces each letter to be a permutation or a rank-$(n-1)$ map with common kernel 01. Pair 02 is new at level three, so some letter f must send it to a distinct pair in P_2. It cannot send it to 01, since that would put 02 in P_2. Thus $02f = 23$ is necessary.

If f is a permutation, its inverse pair of 01 is either 01 or 23 by the definition of P_2. Its inverse pair of 23 would have to be 02. Inverse pairs of the disjoint target pairs 01 and 23 must be disjoint, but 02 intersects both possible inverse pairs of 01. This is impossible.

If f is singular, its common kernel is 01. For $02f = 23$, $f(0)$ is an endpoint of 23 and $f(2)$ the other endpoint. Since $f(1) = f(0)$, also $12f = 23$. This would put the additional pair 12 in P_3, contradicting the three specified pairs. The two cases exhaust every letter and exclude this type.

A.8 Type 6: an edge attached to the kernel pair

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 23\}, \quad P_3 = \{01, 23, 04\}.$$

Let $C = \{0, 1, 2, 3, 4\}$. The three-hole P_3 covers form H: core holes 012,013,023,024,034,124,134 and outside holes $02x$,$03x$, x outside C. Every image reached within six letters has holes in H.

Complete letter classification Each letter is a permutation or a rank-$(n-1)$ map with common kernel 01. Its inverse pairs of 01 belong to $\{01, 23\}$, and inverse pairs of 23 belong to $\{01, 23, 04\}$. Permutations have precisely three forms:

- A: preserve each pair 01,23;
- B: exchange those two pairs;
- T: map 23 onto 01 and 04 onto 23, with the image of state 1 outside $\{0, 1, 2, 3\}$.

The first two have core images in $\{0, 1, 2, 3\}$ and arbitrary outside images. In T the four images of 0,2,3,4 are all of $\{0, 1, 2, 3\}$.

A singular letter cannot have its kernel image in a target pair whose inverse pairs are required: that would give both $0x$ and $1x$, which cannot both lie in either allowed early-pair family. If its missing output is outside 01, it must map 23 onto 01. Its missing output must then be 2 or 3, since otherwise its inverse pair of 23 could be none of the permitted pairs. If it omits 0 or 1, it must map 23 onto 23. Thus all singular letters have one of these forms:

- U: missing e in $\{0, 1\}$, map 23 onto 23, kernel image c in $\{1 - e\} \cup \{4, 5, \ldots\}$;
- V: missing e in $\{2, 3\}$, map 23 onto 01, kernel image c in $\{5 - e\} \cup \{4, 5, \ldots\}$.

Their images of other states are distinct and omit e and c. Only T can create the new pair 04, so some T permutation is necessary. Swapping labels 2 and 3, the chosen T has one of the normalized core rows

$$\text{I} : (2, a, 0, 1, 3), \qquad \text{II} : (2, a, 1, 0, 3), \qquad a \geq 4.$$

Call it closed when $a = 4$ and mixing otherwise. The swap fixes 0,1,4 and preserves the ordered early-pair type.

Orientation II If $a \geq 5$, put $v = T^{-1}(4)$. The common domain for T and TT is exactly

$$\begin{cases} \{013, 034, 134\}, & a = v, \\ \{034, 134, 03w\}, & a \neq v, \quad w = T^{-1}(v) \notin C. \end{cases}$$

For example the one-step domain is 013,023,034,134,02v and all outside 03x. The second test removes 023 and 02v; it retains 013 only if $a = v$, and an outside 03x only if $Tx = v$. Since $T_1 = a$, exactly one of these last possibilities exists. The domain has size three, fewer than $|R_4| \geq 5$, so mixing orientation II is impossible.

If $a = 4$, the T,TT domain is precisely the five core holes

$$D = \{012, 023, 034, 124, 134\}.$$

Its occupied rows form the type-6 five-cycle listed above. The common-entry argument excludes closed orientation II.

Orientation I requires a breaker Let $T = (2, a, 0, 1, 3)$ on C, put $v = T^{-1}(4)$, and $\sigma = [a = v]$. The domain for T,TT is exactly

$$\begin{cases} \{012, 013, 023, 024, 134\} \cup \{02x : x \notin C\}, & T \text{ closed}, \\ \{012, 023, 024, 03v\} \cup \{02x : x \notin C\} \cup \{124 : \sigma = 1\}, & T \text{ mixing}. \end{cases}$$

In the closed case $v = 1$. In the mixing case v is outside C. These identities follow by applying T to each of the seven core holes and to 02x,03x separately; the one-step mixing domain is $\{012, 023, 024, 124, 03v\} \cup \{02x : x \notin C\}$.

For a permutation g, call it a keeper when $\{g(0), g(2)\} = \{0, 2\}$. For a singular g with missing output e, call it a keeper when $\{e, g(2)\} = \{0, 2\}$. If every letter were a keeper, the family of holes 02x for all $x \neq$ 0,2 (including $x =$ 1,3,4) would be invariant. For singular letters, the hole 012 maps to 02c, and every other 02x maps to 02$g(x)$; each is again three distinct holes. Thus these $(n-3)$-images could never compress, contradicting synchronization. There is a breaker g.

All orientation-II letters have already been excluded, so the complete breaker list is as follows. Below h denotes $g(4)$, except in the T row where $h = g(1)$; each free output obeys the rank and missing-output rules.

Name	Core row or conditions
A_{03}	$(0, 1, 3, 2, h)$
B_{03}	$(3, 2, 0, 1, h)$
A_{bad}	$(1, 0, k, l, h), \{k, l\} = \{2, 3\}$
B_{bad}	$(k, l, 1, 0, h), \{k, l\} = \{2, 3\}$
T_{other}	$(3, h, 1, 0, 2), h \geq 4$
U_{03}	$(c, c, 3, 2, h)$, missing 0
V_{03}	$(c, c, 0, 1, h)$, missing 3
U_{bad}	U missing 1
V_{bad}	V with $g(2) = 1, g(3) = 0$

The list follows from the A,B,T,U,V classification by checking whether the carrier of the holes 02x is 02,03,12 or 13. Carrier 02 is precisely the keeper case. A carrier 12 or 13 covers P_3 when its third point is 0, as well as when it is 4; both possibilities are included below.

Two-step bounds Let $D(T, g)$ be the H-domain of all words of length at most two on T,g. Its core holes are among the displayed T,TT core holes. Its outside holes are only $02x$ and, in the mixing case, at most the single $03v$. Write $C_i = [c = i]$, $H_i = [h = i]$. Each triple in the tables bounds

(N_c, N_{02}, N_{03}), where the components count core holes, outside holes of type $02x$, and outside holes of type $03x$, respectively.

For closed orientation I, the bounds are as follows. In the U_{bad} row, write $C = C_0 + C_4$ and $H = H_0 + H_4$.

Breaker	Component bounds
A_{03}	$(4, 0, 0)$
B_{03}	$(3, 0, 0)$
$A_{\text{bad}}, B_{\text{bad}}$	$(3 + H_4, 1 - H_4, 0)$
T_{other}	$(2 + H_4, 1 - H_4, 0)$
U_{03}	$(3 + C_1 + H_1, 1 - C_1 - H_1, 0)$
V_{03}	$(1 + C_2 + H_2 + H_4, 1 - C_2 - H_2, 0)$
U_{bad}	$(2C + 2H, 2 - C - H, 0)$
V_{bad}	$(3 + C_4 + H_4, 1 - C_4 - H_4, 0)$

Every sum is at most four. For U_{bad}, c,h are distinct so C,H are each indicators, giving sum $2+C + H \leq 4$.

For mixing orientation I, use $C = C_2 + C_v$, $H = H_2 + H_v$ in the V_{03} row, and $C = C_0 + C_4$, $H = H_0 + H_4$ in the U_{bad} row. The resulting bounds are:

Breaker	Component bounds
A_{03}	$(1 + H_v + \sigma H_4, 1 - H_v, H_4)$
B_{03}	$(1 + H_v + \sigma, 1 - H_v, 0)$
A_{bad}	$(1 + H_4 + \sigma(1 - H_4), 1 - H_4, 1 - H_4)$
B_{bad}	$(1 + H_4(1 + \sigma), 1 - H_4, H_4)$
T_{other}	$(\sigma H_4, 1 - H_4, H_4)$
U_{03}	$(1 + C_v + H_v(1 + \sigma), 1 - C_v - H_v, 1)$
V_{03}	$(C + H(1 + \sigma), 2 - C - H, 0)$
U_{bad}	$(C + H(1 + \sigma), 2 - C - H, 1)$
V_{bad}	$(1 + C_4 + H_4(1 + \sigma), 1 - C_4 - H_4, H_4)$

Here $v \geq 5$, so the indicator sums in the V_{03} row are again indicators. Every row sums to at most four (some have the sharper bound three).

To verify the table algebraically, apply the hole formula in Section A.1 to the core holes and the two outside types. The restrictions on outside $02x$ reduce to unique preimages of these target sets:

- permutation bad rows: $\{4\}$;
- U_{03}: $\{v\}$, with $v = 1$ in the closed case;
- V_{03}: $\{2, v\}$, with target 1 already having core preimage 3 when closed;
- U_{bad}: $\{0, 4\}$;

- V_{bad}: $\{4\}$, since output 0 already has core preimage 3.

Every target occupied by kernel image c or core image h has no outside preimage. This gives all the negative terms in the second column. For permutation A_{03}/B_{03} the targets of gT on $02x$ are $\{2, v\}$, but output 2 already has a core preimage, leaving only v; when closed, output $v = 1$ also has a core preimage, leaving none.

Core counts are obtained by substituting the five/four candidate core holes of T,TT into g,gT,Tg,gg. For illustration, U_{03} has three possible unconditional closed core holes 013,023,134 and the two extra holes 012,024 only if $c = 1$,$h = 1$. In the mixing case it has unconditional 023, 012 only if $c = v$, and 024 (also 124 if $\sigma = 1$) only if $h = v$.

For V_{03} the closed hole 013 cannot survive: its first image would require $c = 4$ and be 134, but its second image then requires $h = 4$, contradicting distinct outputs. The remaining core candidates are 023,012 if $c = 2$, 024 if $h = 2$ and 134 if $h = 4$. In the mixing case 023 fails gT, and $03v$ also fails gT: its possible first images 013 or 134 both fail T. The displayed C,H restrictions on 012,024,124 then give its table row. The bad U core counts simply test whether c,h belong to $\{0, 4\}$; the bad V rows have three unconditional possible closed core holes, with 012/024 requiring $c = 4$,$h = 4$. In the mixing bad V case only 023 is unconditional, 012 requires $c = 4$, and 024, the optional 124 and $03v$ require $h = 4$. The permutation rows follow from their explicit core templates in exactly the same substitution.

These substitutions establish the component bounds in both tables.

It follows that $|D(T, g)| \leq 4$. But every image reached within four letters belongs to it, whereas $|R_4| \geq 5$. This contradiction excludes orientation I and completes this case.

A.9 Type 7: an edge attached to the second pair

The pair layers are

$$P_1 = \{01\}, \quad P_2 = \{01, 23\}, \quad P_3 = \{01, 23, 24\}.$$

Put $C = \{0, 1, 2, 3, 4\}$. The three-hole covers H of P_3 are the seven core holes 012, 023, 024, 123, 124, 034, 134 and outside holes $02x$,$12x$, x outside C.

Complete letter classification Every letter is a permutation or a rank-$(n-1)$ map with common kernel 01. Inverse pairs of 01 must be 01 or 23; inverse pairs of 23 must be 01,23, or 24. The compatible permutation core rows are precisely

$$A : (k_0, k_1, l_0, l_1, h), \quad B : (l_0, l_1, k_0, k_1, h), \quad T : (k_0, k_1, l_0, h, l_1).$$

where (k_0,k_1) orders 0,1, (l_0,l_1) orders 2,3, and $h \geq 4$. Their outside outputs are distinct and avoid the four specified outputs.

For a singular letter, write e for its omitted output and $c = f(0) = f(1)$. If e is outside 01, the inverse pair of 01 must be 23. Its kernel image cannot lie in 01, since then both $0z$ and $1z$ would be inverse pairs. If e were also outside 23, its inverse pair of 23 would be disjoint from 23, hence could only be 01. That would put c in 23 and give two forbidden inverse pairs there. Therefore e is in 23. If e is in 01, the inverse pair of 23 is 23 or 24, with c outside 23. This gives precisely

$$\begin{array}{ccc} U_A & (c, c, l_0, l_1, h) & e \in \{0, 1\}, \\ U_T & (c, c, l_0, h, l_1) & e \in \{0, 1\}, \\ V & (c, c, k_0, k_1, h) & e \in \{2, 3\}. \end{array}$$

For U, c is 1-e or at least 4; for V, c is 5-e or at least 4. The other images are distinct and avoid e,c and the specified pair. These conditions also suffice for compatibility.

The seven forced domains The following are hole sets, not occupied core sets. An outside point v or w is relabeled 5.

name	hole domain	contains
$D_{6,\mathrm{cl}}$	$023, 123, 024, 124, 034, 134$	R_5
$D_{6,T}$	$024, 124, 034, 134, 02v, 12v$	R_5
$D_{6,A}$	$023, 123, 034, 134, 02v, 12v$	R_5
C_A	$012, 024, 124, 034, 134$	R_4
C_T	$012, 023, 124, 034, 134$	R_4
C_V	$023, 024, 124, 034, 134$	R_4
M_w	$024, 124, 034, 134, 02w$	R_4

Taking complements in the distinguished set gives the occupied families above. All except C_T are excluded by rank at most four; C_T is excluded by the five-cycle common-entry argument. The following substitutions derive the domains exhaustively.

Eliminate all letters taking 2 to 3 For a permutation f with $f(2) = 3$, its free core output is $f(4)$ in form A, or $f(3)$ in form T. If that output is 4, the one-step H-preserving domain is $D_{6,\mathrm{cl}}$. Otherwise $v = f^{-1}(4)$ is outside C, and the domain is $D_{6,A}$ or $D_{6,T}$ respectively. These are checked by mapping the seven core holes and the families $02x$,$12x$ directly.

For a singular U row with $f(2) = 3$ the domain size is $6 - [c = 4]$. To see the count, let z be its free core output. Hole 012 survives exactly when $c = 4$; the two holes 023,123 (form A) or 024,124 (form T) survive exactly when $z = 4$. The four other core holes survive. An outside preimage of 4 supplies two outside holes unless 4 is already c or z. Since c and z are distinct, the count is

$$[c = 4] + 2[z = 4] + 4 + 2(1 - [c = 4] - [z = 4]) = 6 - [c = 4].$$

If $c = 4$, size five contradicts $|R_5| \geq 6$. Otherwise it is precisely $D_{6,\mathrm{cl}}$ when $z = 4$, or $D_{6,A}/D_{6,T}$ with $v = f^{-1}(4)$ outside C. The six-member low-rank families therefore exclude every letter with $f(2) = 3$.

Eliminate V rows omitting 3 For V missing 3 put $A = [c \in \{2, 4\}]$, $B = [h \in \{2, 4\}]$. Its one-step domain has size 4-A+2B: hole 012 contributes A, holes 023,123 fail, the other four core holes contribute 4B, and the outside holes contribute 2(2-A-B). If $B = 0$ it has at most four members. If $B = 1$ and $A = 1$ it has five. The only remaining case has six members, $c \geq 5$ and h in $\{2, 4\}$, and is $D_{6,T}$ with v the outside preimage of the other member of $\{2, 4\}$. It too is excluded.

Eliminate B four-cycles After the preceding cuts a B permutation has $B^2(2) = 2$ or 3. In the latter case a simultaneous exchange of labels 0,1 normalizes it to

$$b = (2, 3, 1, 0, a), \qquad a \geq 4.$$

For $a = 4$ the b,bb domain is C_V, and its low-rank family excludes this case. For $a \geq 5$ put $v = b^{-1}(4)$. The domain is

$$\begin{cases} \{023, 034, 024\}, & a = v, \\ \{023, 034, 02w\}, & a \neq v, \quad w = b^{-1}(v) \notin C. \end{cases}$$

It has only three members, contradicting $|R_4| \geq 5$.

A **B two-cycle and a breaker are necessary** If no B permutation remains, every letter preserves the family I of three-hole sets containing 2 and at least one of 0,1. The remaining A,T permutations fix 2 and preserve 01; U maps the two carriers to $\{e, 2\}$; V omitting 2 maps them to $\{2, f(2)\}$. The both-kernel-endpoints case also preserves I. Images with these holes never compress, contrary to synchronization. Choose a B two-cycle and normalize it by exchanging 0,1 to

$$b = (2, 3, 0, 1, a), \qquad a \geq 4.$$

Call a permutation a keeper if $\{g(0), g(2)\} = \{0, 2\}$, and a singular row a keeper if $\{e, g(2)\} = \{0, 2\}$. Every keeper preserves the family F of all holes $02x$, $x \neq 0,2$. Thus synchronization requires a breaker. After the preceding cuts the exhaustive breaker forms are

$$\begin{array}{lll} P_A & (1, 0, 2, 3, h) & \\ P_T & (1, 0, 2, h, 3) & \\ P_B & (3, 2, 1, 0, h) & \\ U_A & (c, c, 2, 3, h) & e = 1, \\ U_T & (c, c, 2, h, 3) & e = 1, \\ V & (c, c, 1, 0, h) & e = 2. \end{array}$$

Here $h \geq 4$ for permutations; c is 0 or ≥ 4 for U and 3 or ≥ 4 for V. In singular rows h avoids the missing output, c, and the specified pair.

Mixing B two-cycles If $a \geq 5$ put $v = b^{-1}(4)$. Directly testing b and bb gives the domain

$$F \cup \{12v\} \cup \{034 : a = v\}.$$

In particular it has at most two members outside F. For a breaker g, write N_c for the count of core holes in $D(b, g)$ intersect F, and N_o for the count of its outside holes. The following upper bounds follow by mapping the carriers of $02x$ under g and then under b. For U set $C = [c \in \{0, v\}]$, $H = [h \in \{0, v\}]$; for V set $C = [c = v]$, $H = [h = v]$; for P_A,P_T set $H = [h = v]$.

breaker	N_c	N_o
P_A, P_T	$1 + H$	$1 - H$
P_B	0	$1 - [h = 4]$
U_A, U_T	$C + H$	$2 - C - H$
V	$1 + C + H$	$1 - C - H$

For example U converts the carriers to 12; after b these are 03. The third output must be 0 or v so that its b-image is 2 or 4. The outputs already occupied by c or h remove precisely $C + H$ available outside preimages. The analogous V carrier is 12 with output 0 already on input 3, leaving only the target v. The bounds sum to at most two; adding the at most two non-F members gives $|D(b, g)| \leq 4$, contradicting $|R_4| \geq 5$. Hence $a = 4$.

Closed B two-cycles For $a = 4$ the b,bb domain is $F \cup \{124, 034, 134\}$. Count core and outside $02x$ members of $D(b, g)$ as N_c, N_o. For U put $C = [c \in \{0, 4\}]$, $H = [h \in \{0, 4\}]$, $H_0 = [h = 0]$, $H_4 = [h = 4]$; for V put $C_4 = [c = 4]$, $H_4 = [h = 4]$.

breaker	N_c	N_o
P_A, P_T	$1 + 4H_4$	$1 - H_4$
P_B	$1 + 3H_4$	$1 - H_4$
U_A	$C + 2H_0 + 4H_4$	$2 - C - H$
U_T	$C + H_0 + 3H_4$	$2 - C - H$
V	$3 + C_4 + 2H_4$	$1 - C_4 - H_4$

These bounds are obtained by testing the six core holes individually and counting outside preimages of the same carrier targets. Their sums are at most five; P_B and U_T have sums at most four. For U the cases $h = 0$ and $h = 4$ must be distinguished: $h = 0$ does not preserve 034,134.

Since $|R_4| \geq 5$, equality is necessary. Testing its core holes gives:

- P_A has $h = 4$ and domain C_A;
- P_T has $h = 4$ and domain C_T;
- U_A has $h = 4$. If $c = 0$ its domain is C_A; otherwise $c \geq 5$ and its domain is M_w, where $w = g^{-1}(0)$ is outside C;
- V has $h = 4$ and domain C_V.

All these five-member domains are excluded by the rank and common-entry arguments above. This excludes the remaining letters.

A.10 Type 8: three disjoint pairs

Here every safe image selects one vertex from each of $01, 23, 45$ and contains every state outside those six. The three block-coordinate sums are equal, and each outside coordinate equals their common sum. The row space has dimension at most four and contains no reset row $|S|e_q$ (also when $n = 6$ and the outside set is empty). The reset-row version of the plateau argument thus gives $\dim V_4 \geq 5$, whereas R_4 lies in this space. This is a contradiction. The nine cases exhaust the possible early pair layers, proving Theorem 27.

B Structural lemmas

For a graph G, let $N_G(X)$ be the open neighborhood of X and put $N_I(X) = N_G(X) \cap I$. The support of G is its set of nonisolated vertices. We first establish the graph-theoretic tools used in the reduction, and then prove the statements from Section 4.

Lemma 28 (Support bound from near-maximum independent sets)**.** *Suppose every vertex of G lies in an independent set of size at least $\alpha(G) - d$, where $d \geq 0$, and let $t = \tau(G)$. The number N of nonisolated vertices satisfies $N \leq t(d + 2)$ (the support inequality of Gitler–Valencia [3]). If $d > 0$ and $\delta = t(d + 2) - N$, the number of maximum independent sets is at most $2^{\lfloor \delta/d \rfloor}$.*

Proof. Fix a maximum independent set I and put $C = V(G) \setminus I$. For each $c \in C$, choose an independent set of size at least $|I| - d$ containing c, and let X_c be its intersection with C. Then $|N_I(X_c)| \le |X_c| + d$. For any independent $Y \subseteq C$, maximality of I gives $|N_I(Y)| \ge |Y|$. If A is a union of previously chosen X_c, neighborhood submodularity and independence of $A \cap X_c$ give

$$|N_I(A \cup X_c)| - |A \cup X_c| \le |N_I(A)| - |A| + d.$$

At most t choices cover C, so $|N_I(C)| \le t + td$. Every vertex of C is nonisolated, and the nonisolated vertices of I are precisely $N_I(C)$, proving the first assertion.

Every maximum independent set has the unique form $(I \setminus N_I(X)) \cup X$, where $X \subseteq C$ is independent and $|N_I(X)| = |X|$. Let U be the union of these tight exchange sets. The same submodularity argument with $d = 0$ gives $|N_I(U)| - |U| \le 0$. Add at most $t - |U|$ of the sets X_c to cover C. Thus $N - 2t \le (t - |U|)d$, or $|U|d \le \delta$. The maximum sets are indexed by subsets of U. □

Lemma 29 (Neighborhood bound for an independent set)**.** *Under the hypotheses of Lemma 28, every independent set I containing no isolated vertex satisfies $|I| \le (d+1)|N_G(I)|$.*

Proof. Suppose $|I| > (d+1)|A|$, where $A = N_G(I)$. If the bipartite graph between A and I has no matching saturating A, choose $Y \subseteq A$ with $|N_I(Y)| < |Y|$. Replace I by $I \setminus N_I(Y)$ and A by its actual neighborhood, a subset of $A \setminus Y$. The strict inequality persists. Repetition terminates with a nonempty violating pair admitting a matching saturating A; a nonempty set of originally nonisolated vertices cannot have empty actual neighborhood.

There is a maximum independent set R containing this I. Indeed, in any maximum set T, replace $T \cap (I \cup A)$ by I. The matching assigns distinct points of $I \setminus T$ to $T \cap A$, so the replacement preserves maximal cardinality and is independent. Put $C = V(G) \setminus R$. For each $a \in A \subseteq C$, choose a near-maximum set containing a and write its intersection with C as $X_a = Y_a \sqcup Z_a$, with $Y_a \subseteq A$ and $Z_a \cap A = \varnothing$. Since Z_a has no neighbors in I, and $|N_R(Z_a)| \ge |Z_a|$,

$$|N_I(Y_a)| + |Z_a| \le |N_R(X_a)| \le |X_a| + d, \qquad |N_I(Y_a)| \le |Y_a| + d.$$

The Y_a cover A. The saturating matching ensures $|N_I(Y)| \ge |Y|$ for every $Y \subseteq A$. Neighborhood submodularity, adding at most $|A|$ sets Y_a, now gives $|I| = |N_I(A)| \le (d+1)|A|$, a contradiction. □

Lemma 30 (Intersection and union of maximum independent sets)**.** *Let $\mathcal{D}$ be all maximum independent sets of a graph H, all of size s. Put $I_0 = \bigcap \mathcal{D}$, $U_0 = \bigcup \mathcal{D}$, and $B = U_0 \setminus I_0$. Then $|I_0| + |U_0| \ge 2s$. If equality holds, $H[B]$ has a perfect matching, and each member of $\mathcal{D}$ selects exactly one endpoint of each matching edge, together with all of I_0.*

Proof. Fix $R \in \mathcal{D}$. Every member is $(R \setminus N_R(X)) \cup X$ for a tight independent exchange $X \subseteq V(H) \setminus R$, with $|N_R(X)| = |X|$. Let A be the union of the tight exchanges and $D = N_R(A)$. Submodularity as in Lemma 28 gives $|D| \le |A|$. Since $I_0 = R \setminus D$ and $U_0 = R \cup A$, the inequality follows.

In the equality case $|D| = |A|$. Process the tight exchanges successively. Their intermediate union A' has nonpositive neighborhood deficit $|N_R(A')| - |A'|$, which cannot increase; its final value is zero. Consequently every intermediate deficit is zero. When adding X, equality forces

$$N_R(A') \cap N_R(X) = N_R(A' \cap X), \qquad |N_R(A' \cap X)| = |A' \cap X|.$$

Every subset of the independent set X satisfies Hall's inequality, by maximality of R. A matching of X onto $N_R(X)$ therefore exists, and it matches $A' \cap X$ onto its equal-size neighborhood. Its remaining edges extend any already constructed matching of A' onto $N_R(A')$. Induction gives a matching of A onto D. These are the two halves of B. Every maximum set has $|A|$ points in B and at most one on each matching edge, proving the assertion. □

Lemma 31 (Saturated transversal support)**.** *Let $U \subsetneq Q$ be partitioned into $s \geq 2$ nonempty blocks, let $\mathcal{T}$ be their transversal family, and put $k = |U| - s \geq 1$. In a strongly connected synchronizing automaton, fix $S \in \mathcal{T}$ and suppose $Sw \in \mathcal{T}$ for every word w with $|w| \leq k$. There is a state $z \in U$ such that every transition from U to its complement has source z. Every transversal avoiding z maps to a transversal under every letter. The shortest path from S to z has length k, and z lies in a nonsingleton block.*

Write $a = |Q \setminus U|$, $m = k + a$, and let J consist of the singleton blocks. If $c(S) \geq h + m + 1$, then

$$\sum_{v \notin U} |N_{P_h}(v) \cap J| \leq a(a+1)/2.$$

Proof. The reachable unions $U_i = \bigcup_{|w| \leq i} Sw$ grow strictly while proper. Since $|U_k| \leq |U| = s + k$, each of the first k steps adds exactly one state, and $U_k = U$. Write z for the last added state. Every transition from U_{k-1} already lands in U, so every exit from U has source z. Every short image contains every singleton block, hence z is not such a block.

Let W consist of rows supported on U with equal sums on all s blocks. It has dimension $k+1$, and $W_z = W \cap \{x_z = 0\}$ has dimension k. The space V_{k-1} of Lemma 13 lies in W_z and has dimension at least k, hence equals W_z. For any letter f, $V_{k-1} D_f \subseteq V_k \subseteq W$. Thus the row of every transversal avoiding z maps into W. Its nonnegative integer coordinates have total s, so each block sum is one and its image is again a transversal.

For $v \notin U$, let $\rho(v)$ be its distance from z. A shortest such path stays outside U after its first step: returning to U would require another passage through z before exiting, and that cycle could be removed. Thus $\rho(v) \leq a$. For an uncompressed s-set T, put $\varepsilon(T) = \min_{X \in \mathcal{T}} |T \setminus X|$. Changing at most one point in a transversal makes it avoid z. Since function images do not increase one-sided set difference,

$$|Tf| = s \quad \Longrightarrow \quad \varepsilon(Tf) \leq \varepsilon(T) + 1.$$

Reach an actual transversal containing z in k steps and then follow the $\rho(v)$-step path. If compression has not occurred, the resulting image contains v and omits at most $\rho(v)$ points of J. More than $\rho(v)$ neighbors of v in J would therefore produce a P_h edge and compression within $k + \rho(v) + h \leq m + h$, a contradiction. Hence $|N_{P_h}(v) \cap J| \leq \rho(v)$. The positive distance layers are consecutive; on a vertices their sum is at most $1 + \cdots + a$, proving the stated bound. □

Proof of Lemma 11

Proof. Let $K_1, \ldots, K_k$ be the kernel blocks of u, and put $d = n - k + 1$. Assume all words of length at most $a + d$ preserve the cardinality of T. For every $|v| \leq d$, the k trajectories of Tv visit the k blocks once each; otherwise vu compresses T within $a + d$ letters. Consequently the characteristic row of Tv lies in the linear subspace

$$H = \{x \in \mathbb{R}^Q : \sum_{q \in K_1} x_q = \cdots = \sum_{q \in K_k} x_q\}, \qquad \dim H = n - k + 1 = d.$$

Let $V_i = \operatorname{span}\{\chi_T M_v : |v| \leq i\}$. A plateau $V_i = V_{i+1}$ makes V_i invariant under every letter. It then contains the row obtained from a reset word, which has mass k concentrated in a single kernel block; that row lies outside H. Hence $\dim V_0 < \dim V_1 < \cdots < \dim V_d$, yielding $\dim V_d \geq d + 1$, a contradiction. □

Proof of Lemma 12

Proof. The case $d > m$ follows from the cardinality of Qu. When $d = m$, Lemma 11 applies to the $(n-m)$-set S and gives $c(S) \le a + m + 1$. For $d < m$, the proof of Lemma 11, applied to $T = Qu$, supplies a prefix v of length at most $d+1$ for which vu compresses Qu. Thus $Q(uvu)$ has rank at most $n-d-1$. The word uvu has length at most $2a+d+1$. Induction on $m-d$, together with monotonicity of the right-hand expression in the attained defect, gives

$$c(S) \le 2^{m-d-1}(2a+2d+4) - 1 = 2^{m-d}(a+d+2) - 1.$$

Every concatenation counts its full length. □

Proof of Lemma 13

Proof. A plateau $V_i = V_{i+1}$ makes the space invariant under every letter. A reset word followed by a path to any state q supplies the row $|S|e_q$, so the eventual span is $\mathbb{R}^Q$. Starting with dimension one, the dimensions strictly increase until n. □

Proof of Lemma 14

Proof. All depth-m images are independent in G and cover Q. Every vertex of G therefore belongs to an independent set of size $n - m = \alpha(G) - d$. All depth-$(m-\ell)$ images belong to $\mathcal{D}$; their union has at least $n - \ell$ points, so $a \le \ell$. Put $B = (\bigcup \mathcal{D}) \setminus I_0$ and $p = |K_h \cap I_0|$. Lemma 30 gives $|B| \le 2(m-a)$. Every G-neighbor of I_0 belongs to A_0. The independent-neighborhood ratio gives $p \le a(d+1)$. Thus $|W| = a + |B| + p \le 2m + ad$.

For the strict bound it remains to consider $a = \ell$ and $|B| = 2(m-a)$, since either strict inequality saves at least one. Lemma 30 provides a perfect matching on B. Partition $U = Q \setminus A_0$ into its matching edges and the singletons I_0. The family $\mathcal{D}$ is contained in the transversals of this partition, whose defect is $k = m - \ell \ge 1$. All depth-k actual images are transversals. Apply Lemma 31: $p \le a(a+1)/2$ because each nonisolated point of I_0 has a neighbor in A_0. Consequently

$$|W| \le 2m + \ell(\ell-1)/2 \le 2m + \ell d - 1 \quad (1 \le \ell \le 2d).$$

Finally the complement of each row is a cover of size m; at least t of its vertices are nonisolated in G. □

Proof of Lemma 15

Proof. The depth-m images are independent in P_h and cover Q. An independent $(n-m)$-set containing a vertex excludes all its neighbors, giving the degree bound. Lemma 28 with $d = m - t$ gives the support bound. A short-pair layer cannot plateau before becoming complete: its recurrence closes under one-letter preimages and direct kernels, and a plateau would prevent any remaining pair from ever merging. Since S is independent and has at least two points, the first h layers are proper and contribute at least one new edge each.

For the upper edge bound, consider any graph of maximum degree at most m and cover number at most m with fewer than $3m$ edges. It admits an $(n-m)$-set inducing exactly one edge. Indeed, take a minimum cover C of size t' and its independent complement I. If $c \in C$ has $d_I(c) \le m - t' + 2$, retain one of its neighbors, delete the others from I, and add c, then trim to size $n - m$. If no such c exists, $t' \le 2$ contradicts the degree bound; for $3 \le t' \le m$ there are at least $t'(m - t' + 3) \ge 3m$ edges, again a contradiction. Deleting the one induced edge and repeating gives a Frankl–Pin

sequence using every edge: no later edge lies in an earlier chosen set. Thus the maximum sequence length p over any graph obeys $p \geq \min(|E|, 3m-1)$.

The Frankl–Pin refinement [4] gives $c(S) \leq B_m - p + h$. For completeness, if a shortest compressing word has length $L' > h$, take its first $L' - h$ prefix images and at each prefix a pair merged by the corresponding suffix. A later selected pair belongs to no earlier image, since otherwise that suffix would shorten the word. None of the images contains a pair merging within h letters. Appending a length-p sequence over P_h gives a Frankl–Pin sequence of length $L' - h + p \leq B_m$. If $|E(P_h)| \geq b+1$, this implies $c(S) \leq B_m - (b+1) + h = L - 1$. Finally depth-$(L-h-1)$ images are maximum independent sets when $t = m$, and Lemma 13 gives their rank bound. □

Proof of Lemma 16

Proof. Lemma 15 first gives $|K_9| \leq 12$. Suppose $|K_9| \geq 11$. Since S is independent in P_{10}, $4 \leq \tau(P_{10}) \leq 5$. If $\tau(P_{10}) = 5$, apply Lemma 14 with $m = 5$, $h = 9$, $\ell = 1$, $d = 1$ to obtain $|W| \leq 10$, contradicting $K_9 \subseteq W$. Thus $\tau(P_{10}) = 4$. Every maximum independent set of P_{10} maps under every letter, injectively, into a maximum independent set of P_9.

In Lemma 28, the parameters $t = 4$, $d = 1$ give deficit $\delta = 12 - |K_9|$. If $|K_9| = 12$, there is a unique maximum set R of P_9. It is also the unique maximum set of P_{10} and satisfies $Rf = R$ for every letter, contradicting strong connectivity. For $|K_9| = 11$, the proof of that lemma gives a tight-exchange union of size at most one. Hence there are at most two maximum sets, and if two exist they differ by one state. A unique maximum set is excluded as above. Denote the two by R, T. If both remained independent in P_{10}, each letter would map both into $\{R, T\}$ and preserve their proper union of size $n-3$, again contradicting strong connectivity. Exactly one remains; denote it by R_0. Every letter maps R_0 into $\{R, T\}$. □

Proof of Lemma 17

Proof. A member of $\mathcal{D}$ is either in $\mathcal{B}$ or is obtained by deleting one point from a member R of $\mathcal{M}$. There are at most two such deletions for a fixed R: three deletions would make every pair of R co-occur in an H-independent set, contrary to $\alpha(H) = s$. This proves the cardinality bound, allowing overcounting between maximum sets.

If a member of $\mathcal{D}$ omits an isolated vertex z of G, adding z gives a member of $\mathcal{M}$. If some R supplies two such rows, $R \setminus \{z\}$ and $R \setminus \{z'\}$, then zz' is an edge of H: it is the only pair of R absent from both rows, and some edge must occur in $H[R]$. Every member of $\mathcal{D}$ must omit z or z'. Its complement, a size-$(|V| - s)$ cover of G, can omit at most one G-isolate since $\tau(G) = |V| - s - 1$. Thus $Z \subseteq \{z, z'\}$. Otherwise each member of $\mathcal{M}$ supplies at most one omitted isolate, giving $|Z| \leq |\mathcal{M}|$. □

Proof of Lemma 18

Proof. More generally, if D is independent in P_H and $|w| \leq H - h$, then $|Dw| = |D|$ and Dw is independent in P_h. A collision under w would merge a pair of D within H letters. A P_h edge in Dw, preceded by w, would do the same. This proves the first and fourth conditions. In the latter case the same letter acts in both positions of its square. The second condition is the recurrence for shortest merging words. For the third, project paths of $\mathcal{A}$ onto the distinguished states and the common reservoir, omitting the reservoir vertex when it is empty. Strong connectivity is preserved under this projection. Finally $c(S) \geq L > H$ makes S independent in P_H; its size $\alpha(P_H)$ places it in $\mathcal{D}$. Using the signatures of the letters and the actual shortest distances gives the asserted selection. □

Proof of Lemma 19

Proof. Apply Lemma 15 with $(m, L, h) = (4, 12, 7)$. It gives degree at most four, support at most eight, seven through ten edges and row rank at least five. Cover number four forces at least five nonisolated vertices. □

Proof of Lemma 20

Proof. The set S is independent in P_9, giving cover number at most five. Theorem 27 applied to all $(n-3)$-sets gives cover number at least four. In the cover-five case, depth-five actual images are maximum independent sets and cover all states. Lemma 15, with $(m, L, h) = (5, 15, 9)$, gives the stated degree, support, edge and rank bounds. Complete generation of graphs of orders six to ten with degrees one through five, followed by the precise cover, edge and row tests, yields the classes in Table 1. The corresponding joint one-letter and distance configurations are handled in Section 5. □

Proof of Lemma 21

Proof. This is the eleven-vertex case of Lemma 16. R_0 is the unique maximum independent set of P_{10}. □

Proof of Lemma 22

Proof. For each neighbor of x in the image Qf, choose a preimage. Together with the isolated preimage of x, it gives a G-nonedge, and its image is a G-edge. At most $d(f)$ neighbors lie outside Qf. The nontrivial fibers of f give at least $d(f)$ distinct kernel pairs in P_1. Every edge counted by $F_G(f)$ has merging distance exactly h: a shorter merge, preceded by f, would merge its nonedge preimage within h letters. The layers $P_i \setminus P_{i-1}$, $2 \le i \le h-1$, are nonempty. Indeed, a pair of merging distance greater than h exists because G is proper; the suffixes of a shortest merging word supply every intermediate distance. These disjoint layers contain at least $d(f) + (h-2) + F_G(f)$ edges in total. □

Proof of Remark 23

Proof. Choose a pair of states with minimal merging distance $d > h$ and follow a shortest merging word. Its successive pair distances decrease by at most one, and the first step from distance $h+1$ reaches distance h. The corresponding letter provides the required entrance. □

C Constructions for the smaller values of q

Theorem 32. *The following bounds hold, and every lower bound has a binary witness.*

1. $\mathcal{F}(n, r+1, r) = n - r$ *for* $n \ge 2$ *and* $1 \le r \le n-1$.
2. $\mathcal{F}(n, r+2, r) = n - r$ *for* $n \ge 4$ *and* $1 \le r \le n-3$.
3. *For* $r \ge 1$, $s - r \ge 3$ *and* $n \ge 2s - r - 1$, *one has*

$$\mathcal{F}(n, s, r) = n - r.$$

For $1 \le r < n$, *one has* $\mathcal{F}(n, n, r) = 1$.

Thus the first two cardinality faces are determined, including their full-image boundary. For every fixed (s, r) with $s - r \geq 3$, the bound is sharp at every $n \geq 2s - r - 1$, in particular at $n = r + k(s - r - 1) + 2$ for $k \geq 2$.

The upper bound follows from Lemma 2; when $S = Q$, a nonpermutation letter compresses immediately.

We will also use the following elementary fact. If t cycles $x_0, x_1, \ldots, x_{s-1}$ and e sends x_0 to x_1 while fixing the other points, then

$$e(t^{s-1}e)^{s-2} \tag{4}$$

maps this set to x_1. Indeed, after e the image is $\{x_1, \ldots, x_{s-1}\}$; each $t^{s-1}e$ deletes the last point of the remaining interval. This also covers $s = 2$.

C.1 The first two cardinality faces

Proposition 33. *Part (1) of Theorem 32 holds.*

Proof. Put $L = n - r \geq 1$ and $Q = \{0, \ldots, n - 1\}$. Define

$$xa = \begin{cases} x + 1 & x < L, \\ x & x \geq L, \end{cases} \qquad S = \{0, L, L + 1, \ldots, n - 1\}, \qquad A = S \setminus \{0\}.$$

Let b cycle S in the displayed order and send every point outside S to L. Then $Qb = S$. The a-path from 0 to L and the b-cycle prove strong connectivity.

For $0 \leq j < L$, set $S_j = A \cup \{j\}$. One has $S_j b = S_0$, and $S_j a = S_{j+1}$ for $j < L - 1$; $S_{L-1}a = A$. Induction shows that every word of length less than L has image S_j with $j < L$, so is safe. The word a^L compresses S, proving $\mathrm{ac}(S, A) = L$.

On S, $c = a^L$ is the unit $0 \mapsto L$. Formula (4) with $t = b$ shows that $c(b^r c)^{r-1}$ resets S. It resets Q as well, since $Qc = A = Sc$. This proves synchronization, also when $L = 1$ or $r = 1$. □

Proposition 34. *Part (2) of Theorem 32 holds.*

Proof. Put $M = n - r \geq 3$. Take disjoint sets $A = \{p_0, \ldots, p_{r-1}\}$ and $U = \{u_1, \ldots, u_M\}$, and let $S = A \cup \{u_1, u_2\}$. Letter a fixes A, while

$$p_i b = p_{i+1} \quad (0 \leq i < r - 1), \qquad p_{r-1} b = u_1.$$

For odd M, let a cycle $u_1, \ldots, u_M$ and set

$$u_j b = \begin{cases} p_0 & j \text{ odd}, \\ u_2 & j \text{ even}. \end{cases}$$

For even M (necessarily $M \geq 4$), set

$$u_1 a = u_2, \quad u_j a = u_{j+1}\ (2 \leq j < M), \quad u_M a = u_2, \qquad u_j b = \begin{cases} u_2 & j = 1 \text{ or } j \geq 3 \text{ odd}, \\ p_0 & j \text{ even}. \end{cases}$$

In both cases $Qb = S$. From p_0 the b-path reaches u_1, whose a-orbit covers U; some state of U returns to p_0 under b. Every state therefore reaches and is reached from p_0.

The two moving points of Sa^j remain distinct for all $j \geq 0$. For $0 \leq j \leq M - 2$, their b-images are p_0 and u_2, so $Sa^j b = S$. At $j = M - 1$, the pair is $\{u_M, u_1\}$ for odd M and $\{u_M, u_2\}$ for even M; both points map to p_0. Thus $h = a^{M-1}b$ compresses S. Every word shorter than M splits

into a-runs followed by b, each run having length at most $M-2$, and a trailing a-run. All these transitions are safe. Hence $\mathrm{ac}(S,A)=M$.

Set $s=r+2$, and $t=ab$ for odd M, $t=b$ for even M. On S, t cycles $p_0,\dots,p_{r-1},u_1,u_2$. If e is the unit $u_1\mapsto u_2$ on S, then $h=et$ there. Consequently

$$h(t^{s-2}h)^{s-2}=e(t^{s-1}e)^{s-2}t$$

on S, and this is a reset by (4). Prepending b gives a reset of Q. The unit e is used only to verify the identity; the reset word uses the letters. □

C.2 Every larger fixed gap

Proposition 35. *For $r\ge 1$, $g\ge 2$ and $M\ge 2g$, there is a binary strongly connected synchronizing automaton with*

$$n=r+M+1,\quad |S|=s=r+g+1,\quad |A|=r,\qquad \mathrm{ac}(S,A)=M+1.$$

The subset S is an actual reachable image.

Proof. Take disjoint sets $P=\{p_0,\dots,p_r\}$ and $U=\{u_1,\dots,u_M\}$. Set $A=P\setminus\{p_r\}$ and $S=P\cup\{u_1,\dots,u_g\}$. First suppose $M\ge 2g+1$ and put $h=1+((M-1)\bmod g)$. Then $1\le h\le g$ and $M\ge 2g+h$. Let a fix P, send u_j to u_{j+1} for $j<M$, and send u_M to u_{h+1}. Define $p_0b=u_h$, $p_ib=p_{i-1}$ for $1\le i\le r$, and

$$u_jb=\begin{cases} p_r & j\equiv h-1 \pmod g,\\ u_1 & j=g,\ h\ge 2,\\ u_{h+g} & j=M,\\ u_{j+1} & \text{otherwise.}\end{cases}$$

The cases are disjoint. On S, b cycles $p_0,u_h,\dots,u_g,u_1,\dots,u_{h-1},p_r,\dots,p_1$. The segment $u_1,\dots,u_{h-1}$ is empty for $h=1$. Ordinary outside states follow increasing indices to the next index congruent to $h-1$, then enter p_r. The exceptional path of length $g+1$ is

$$u_M\longrightarrow u_{h+g}\longrightarrow\cdots\longrightarrow u_{h+2g-1}\longrightarrow p_r.$$

Its indices exist because $M\ge 2g+h$. Thus $Qb^{g+1}=S$. The a-path from u_1 covers U, and the b-cycle covers P, proving strong connectivity.

Safety. Use singleton P-cells and color cells $1,\dots,g$, assigning u_j color $1+((j-h-1)\bmod g)$. Let C be the $n\times s$ state-to-cell incidence matrix. Let T_a fix P and cycle the colors, and let T_b cycle $p_0,g,1,\dots,g-1,p_r,\dots,p_1$. For a standard state column f_q and cell row e_c, one has

$$D_aC=CT_a,\qquad D_bC=CT_b+f_{u_M}(e_g-e_1). \tag{5}$$

The last term is an outer product of an $n\times 1$ column and a $1\times s$ row. These cells are bookkeeping devices, not a quotient automaton. For $w=\ell_1\cdots\ell_t$, iteration gives

$$\chi_SM_wC=\mathbf{1}_s+\sum_{\substack{1\le i\le t\\ \ell_i=b}}\mu_i(e_g-e_1)T_{\ell_{i+1}\cdots\ell_t},\qquad \mu_i=(\chi_SD_{\ell_1\cdots\ell_{i-1}})_{u_M}\ge 0. \tag{6}$$

This counts trajectories even after collisions, with products in word order. A trajectory from S needs at least $M-g$ letters to reach u_M: initial indices are at most g, every transition within

U increases an index by at most one, and entrances from P have index $h \le g$. Hence a nonzero correction has $i \ge M - g + 1$. In the cell graph, color g needs at least $g - 1$ letters to reach color $g - 1$, and color 1 needs at least g letters to reach A. The latter route traverses $1, \dots, g - 1$ and then takes two letters b through p_r. A detour through P cannot shorten either route.

For $t \le M - 1$, correction suffixes have length at most $g - 2$, so no positive correction reaches color $g - 1$, whose trajectory count is at most one. Suppose a word of length at most M has a first collision. For a colliding b, the distinct source states must lie in its only non-singleton fiber, all of color $g - 1$, contradicting this count bound. For a colliding a, the sources must be u_h, u_M, its only non-singleton fiber. They are not both initially present. A preceding b cannot give u_M, which has no b-preimage. A preceding a requires sources u_{h-1}, u_{M-1} one step earlier, both of color $g - 1$, again a contradiction; for $h = 1$, u_h has no a-preimage at all. Thus every word of length at most M preserves s states. For $t \le M$, every correction suffix has length at most $g - 1$, so no negative correction reaches A. Its singleton cells retain multiplicity at least one. Every such word is safe.

The word $a^{M-g}b^{g+1}$ has length $M+1$. Exactly its first b has a trajectory at u_M, with multiplicity one. Its negative correction is $e_1 T_b^g = e_{p_{r-1}}$, whereas its positive correction is $e_g T_b^g = e_{p_r}$. It therefore omits p_{r-1} and attains equality, also when $r = 1$.

For synchronization put $c = a^{M-h}b^s$. The a-run fixes the cyclic tail $u_{h+1}, \dots, u_M$. For $j < h$, it sends u_j to u_{M-h+j}, which has the same remaining b-distance $h - j$ to p_r as u_j. For $j = h$, it sends u_h to u_M, which reaches p_0 in $g + 1 + r = s$ letters b. Consequently c fixes $S \setminus \{u_h\}$ and sends u_h to its b-predecessor p_0. After c, each bc extends the missing interval by one point in forward b-order. Thus $b^{g+1}c(bc)^{s-2}$ resets Q.

The boundary $M = 2g$. Let a fix P and cycle U. Define $p_0 b = u_g$, $p_i b = p_{i-1}$ and

$$u_j b = \begin{cases} p_r & j \equiv g - 1 \pmod g, \\ u_{j+1} & j < M,\ j \not\equiv g - 1 \pmod g, \\ u_M & j = M. \end{cases}$$

Now b cycles $B = P \cup \{u_g, \dots, u_{2g-1}\}$ and fixes u_{2g}; $Qb^{g+1} = E = B \cup \{u_{2g}\}$. Put $d = a^{g-1}b^s$. It fixes P and on E has actions

$$u_g d = u_{2g-1}, \quad u_{g+1} d = u_{2g}, \quad u_{g+j} d = u_{g+j-1}\ (2 \le j \le g - 1), \quad u_{2g} d = u_{2g-1}.$$

Therefore $Eda^g = S$, proving actual reachability. The two cycles prove strong connectivity. Assigning u_j color $1 + ((j - 1) \bmod g)$ gives exactly (5)–(6) again. The same delay and collision arguments apply, with a now a permutation. The word $a^g b^{g+1}$ attains avoidance: only the first negative correction has a suffix of g letters b and reaches p_{r-1}; no positive correction reaches A. Finally $e = d^g$ fixes $E \setminus \{u_g\}$ and sends u_g to u_{2g}. Each be removes one further point of B while retaining u_{2g}. Hence $b^{g+1}e(be)^{s-1}$ resets Q. □

Taking $g = s - r - 1$ and $M = n - r - 1$ proves part (3) of Theorem 32.

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